IMPERIAL

Department of Mechanical Engineering

Advanced Mechanical Engineering Course

Analytical and Computational Investigation of Noise Radiation and Propagation from Lifting Devices

By

Muhammad Abi Rizky

Bachelor of Engineering, Mechanical Engineering, Universitas Indonesia

A thesis submitted in partial fulfilment of the requirements for the Master of Science Degree of Imperial College London and the Diploma of Membership of Imperial College

September 2026

Word Limit: 18000

Word Count: XXXX

Abstract

Open rotor aeroengines offer significant improvements in propulsion efficiency, but expose fan-generated noise directly to the atmosphere. Predicting this noise requires robust acoustic solvers. This report outlines the preliminary development and validation of an analytical Ffowcs Williams-Hawkings (FW-H) code using a permeable surface, retarded-time formulation. The numerical integration logic was evaluated against exact analytical derivations for both monopole and dipole sources under stationary, co-translating, and relative-motion conditions.

The code demonstrated good accuracy ($< 1\%$ error) for monopole sources, in which it captures convective terms and Doppler phase shifting, given sufficient grid and temporal resolutions. For dipole sources, derived analytical solutions were mathematically proven to fail when physical assumptions are violated. To bypass these limitations, a dual-monopole approximation was implemented and validated. This approach is sufficiently robust across all regimes, as it preserves the convective amplification factors that explicit Taylor expansions omit that was required when defining the dipole source term. While some validation work is still necessary, future work will move to a hybrid computational-analytical framework, using Unsteady Reynolds Averaged Navier Stokes (URANS)-FW-H hybrid to predict open rotor noise generation and propagation.

Acknowledgements

LOREM IPSUM

Contents

List of Figures

List of Tables

Nomenclature

Variable	**Definition**
c	speed of sound
M	Mach number ($M = u_i/c$)
p	pressure
ρ	density
u	velocity
t	observer time
τ	retarded time
$\Box^2$	d'Alembertian wave operator, $\Box^2 = \nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}$
∇^2	Laplacian differential operator, $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$
$\delta(f)$ and $\delta(g)$	Dirac delta function of an arbitrary function f and g
$H(f)$	Heaviside step function, $f < 0$ returns $H(f) = 0$, and $f > 0$ returns $H(f) = 1$
δ_{ij}	Kronecker delta, $i = j$, $\delta_{ij} = 1$; $i \neq j$, $\delta_{ij} = 0$
y	static coordinate system
η	moving coordinate system
J	Jacobian of $y \rightarrow \eta$ coordinate change
$q(x,t)$	monopole source strength
$f_i(x,t)$	dipole source strength. Appendix C, arbitrary response.
T_{ij}	Lighthill stress Tensor, $T_{ij} = \rho u_i u_j + p_{ij} - c^2 p' \delta_{ij}$
λ	wavelength
k	wave number, $k = 2\pi/\lambda$ or an integer value for Tyler-Sofrin mode, context provided each time
f	wave frequency
T	wave period, $T = 1/f$
r	radiation distance magnitude
R	control surface radius
δ	small distance between two equal monopoles of opposite signs
n	normal direction vector
m	Tyler-Sofrin spatial mode
n	Tyler-Sofrin temporal mode

Variable	**Definition** (continued)
θ	Azimuthal angle
ϕ	Phase angle
ω	Interaction frequency or source pulsation frequency, context provided each time
Ω	Shaft speed
B	Rotor blade count
V	Stator blade count
d	Circumferential distance between two grid points
N	Discretisation count, e.g., grid count or time-step count
X	Spatial discretisation parameter, number of grid points per wavelength
P	Temporal discretisation parameter, number of time step count per period
ψ	Appendix C, arbitrary wave
Subscripts	
0	value in stationary flow
r	resolved in the direction of radiation
min	minimum/shortest value of interest
max	maximum/largest value of interest
$mono$	monopole source
$dipole$	dipole source
ref	reference value
num	numerical value
i	vector towards the i-direction
j	vector towards the j-direction. Subchapter 2.2 the j-th index of a sum
m	m-th Tyler-Sofrin spatial mode
n	n-th Tyler-Sofrin temporal mode
$fore$	Front blade row
aft	Aft blade row
s	Value at source, e.g., λ_s and f_s mean source wavelength and source frequency, respectively, prior to Doppler-shifting at the observer
Superscripts	
$'$	perturbation component, e.g., $p' = p(x,t) - p_\infty$

Chapter 1

Introduction

1.1 Motivation

Due to environmental concerns and the availability of fossil fuels, the efficiency of civil aero-engines has continuously been improved over the last few decades. Modern engines have increased the bypass ratio to improve efficiency [1], which is supported by increasing the nacelle and fan sizes. The open rotor (also known as ”propfan”) architecture improves on this by placing larger fans outside of the nacelle, potentially further reducing fuel consumption by 10-20% [2] while producing more thrust than existing engines and less carbon emissions.

However, aeroengine nacelles are designed to suppress the fan-generated noise through the use of acoustic linings, so this concept exposes it directly into the atmosphere, causing high noise pollution. An investigation of this noise generation and its propagation is required to develop suppression methods to reduce this noise. In this work, a hybrid approach between the Ffowcs Williams-Hawkings (FW-H) method and computational fluid dynamics (CFD) are used to predict the noise generation and propagation of an open rotor engine.

1.2 Open Rotor Noise Background

Unlike conventional turbofan engines, open rotor architectures lack a nacelle with acoustic liners to suppress the noise. Therefore, the noise generated by the rotors is radiated directly into the atmosphere without suppression. Open rotor noise is caused by the aerodynamic interactions and acoustic interference (inter-rotor and inter-mode interferences) [3]. The reason for this is the acoustic pressure fields generated by rotating blade rows and their aerodynamic interactions. To understand these modal interactions, the Tyler-Sofrin theory for rotor-stator wake interaction is used as the foundation.

The acoustics of a single rotor blade passing a stationary vane can be described as a periodic pressure fluctuation repeating at the blade-passage frequency. At a reference position, this fluctuation is expressed as a Fourier series in time, which can then be expanded into a double Fourier series in space and time to describe the spatial-temporal components of the pressure field. For a single rotor blade-stator interaction, this component is defined as,

$$p'_{m,n} = a_{m,n}\cos(m\theta - nB\Omega t + \phi_{m,n}) \tag{1.1}$$

which represents a traveling wave with m lobes.

When this single-event model is superposed in an array of V evenly spaced stator vanes, the pressure contributions from each vane are shifted in space by the azimuthal angle $\Delta\theta = 2\pi/V$ and in time by $\Delta t = \Delta\theta/\Omega$. Most of these interferences are destructive; however, when $\Delta\theta = \Delta t$, the lobe number m is conserved to satisfy the Tyler-Sofrin mode,

$$m = nB + kV \tag{1.2}$$

where k is any positive or negative integer.

Extending this to the Contra-Rotating Open Rotor (CROR) architecture, this mechanism governs the inter-rotor interaction, with the kinematics being driven by two independently rotating blade rows. By using the aft blade as the reference frame, it acts as a stationary stator where $V = B_{aft}$ vanes spinning at an absolute angular velocity of Ω_{aft}. The front rotor, having B_{fore} blades, interacts with this aft row at a relative angular velocity of $\Omega_{fore} - \Omega_{aft}$. Applying the Tyler-Sofrin condition in this relative frame changes Equation 1.2 into $m = nB_{fore} + kB_{aft}$. Transforming this back to the absolute stationary frame shows the interaction frequency by adding the reference frame's rotation,

$$\omega = nB_{fore}\Omega_{fore} + kB_{aft}\Omega_{aft} \tag{1.3}$$

Consequently, the pattern spin speed of spatial mode m, evaluates to,

$$\Omega_m = \frac{\omega}{m} = \frac{nB_{fore}\Omega_{fore} + kB_{aft}\Omega_{aft}}{nB_{fore} + kB_{aft}} \tag{1.4}$$

and this kinematic relationship is the reason why CRORs generate such loud noises. As the rotors are rotating in opposite directions, Ω_{fore} and Ω_{aft} have opposite signs; therefore, certain harmonics will create combinations of n and k that will cause the denominator in Equation 1.4 to be very small. This increases the pattern spin speed Ω_m significantly. If the spinning mode's tip Mach number is subsonic, the sound intensity decays exponentially with distance from the rotor; however, for the case of CRORs, this spinning mode may go above supersonic, which propagates the sound unattenuated into the far-field [4].

Previous studies have examined the noise data indicating that the frequencies corresponding to shaft orders dominate the sound field [5, 6, 7]. Because the propagation of this noise is governed by the blade counts and rotational speeds, the sound field is sensitive to the rotor's kinematic design. It is found that the dominating noise comes from shaft orders that correspond to the blade passing frequency (BPF) of $1 \times BPF_{aft}$, $1 \times BPF_{fore}$, $BPF_{fore} + BPF_{aft}$, $2 \times BPF_{aft} + 1 \times BPF_{fore}$, and $1 \times BPF_{aft} + 2 \times BPF_{fore}$, which confirms the inter-rotor and inter-mode interferences.

1.3 Hybrid Noise Prediction Background

Predicting open rotor noise requires solving the inhomogeneous wave equation. The FW-H equation provides the standard analytical framework for this process [8]. In this work, the permeable surface formulation (Formulation 1A) is utilised [9, 10, 11]. This formulation was chosen as it captures all noise sources, including quadrupoles (sounds from non-linearities such as shocks and turbulence), via surface integration, provided the control surface fully encapsulates the non-linearities [12], which avoids the computationally expensive volume integrals prohibitive to noise prediction.

In modelling acoustic wave propagation, a few methods are available such as Direct Noise Computation (DNC), FW-H analogy, and Hanson's model, to name a few. It is most practical, however, to conduct the analysis through a hybrid approach, in which flow fields surrounding the noise source are computed using CFD, and the far-field acoustic propagation are computed separately, such as using the FW-H method [13, 14]. This is to avoid the high computational cost of directly resolving the acoustic waves, while having a good geometric sensitivity of the sound source [15, 16, 6], unlike pure analytical approach. The schematic below in Figure 1.1 shows the available methods as well as highlights the current method used in this work.

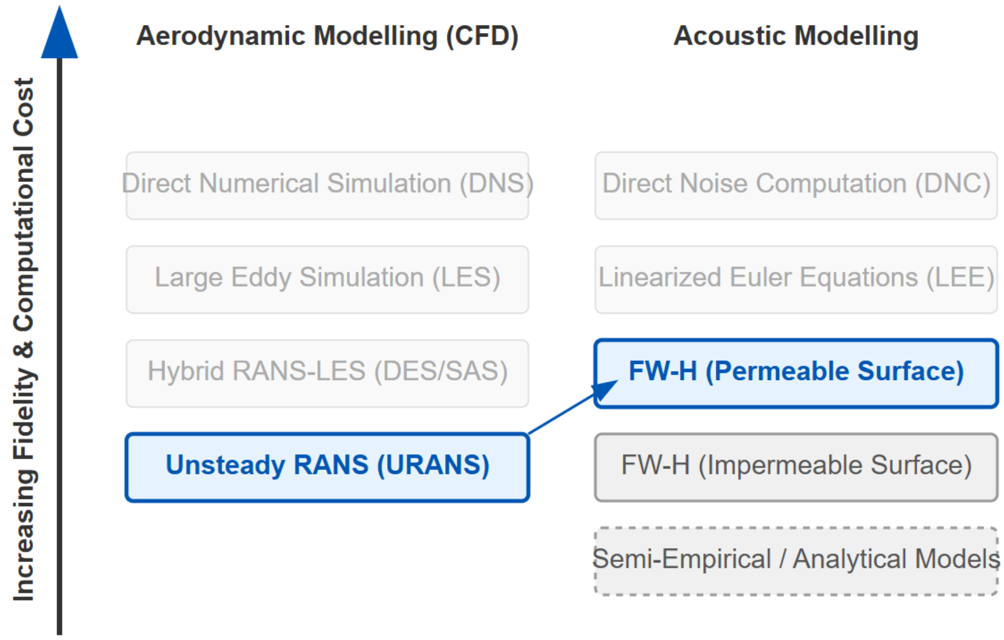


Figure 1.1: Hybrid modelling approach schematic

In this work, the FW-H method is validated and verified for its sensitivity using theoretical point sources, as illustrated in Figure 1.2, to build confidence in its capabilities; the detailed description of which will be provided in the next chapter. After confidence in the tool has been established, CFD will be used as an input to the FW-H code to form a hybrid method [17, 12, 18, 19], which replaces the point source in the illustration below with the CFD domain. As the interest in this work is to predict the tonal noise generated by the open rotor, the CFD framework used is the Unsteady Reynolds Averaged Navier-Stokes (URANS), as it is considered sufficient in providing a good balance between accuracy and computational cost for this purpose [16, 20]. Should a future interest arise in predicting broadband noise, other frameworks such as Detached Eddy Simulation (DES) [21] or Large Eddy Simulations (LES) can be implemented alongside the validated FW-H method.

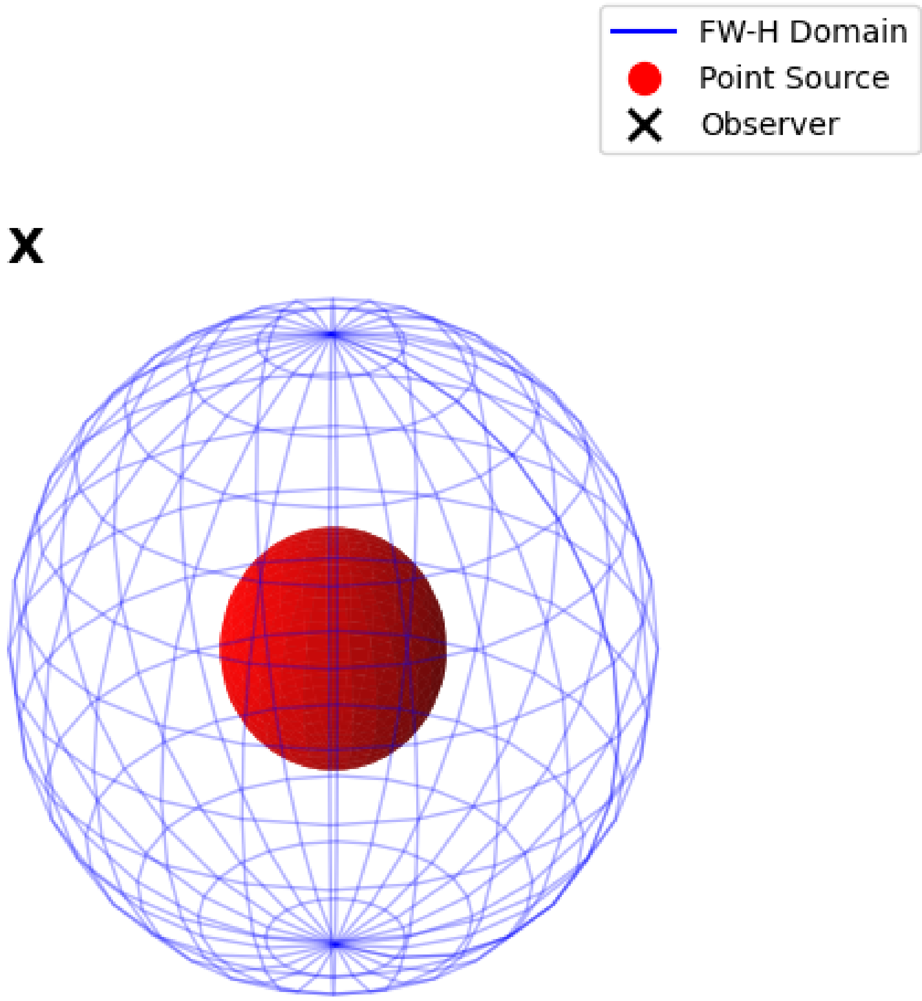


Figure 1.2: Hybrid domain illustration

1.4 Aeroacoustics Prediction using Hybrid Method Background

The prediction of aeroengine aeroacoustics has evolved alongside the advancements in computational fluid dynamics (CFD). After the initial development of the FW-H method [22, 9], between the 1970s and 1980s, aeroengine noise prediction relied almost entirely on analytical and semi-empirical models [23, 24, 3, 25]. These early developments applied the FW-H directly on solid surfaces using theoretical aerodynamic loading distributions. While these models identified the primary mechanisms which caused the noise radiated from open rotors, the theoretical sources used in these studies limited the accuracy of the calculations due to the lack of geometric fidelity that CFD provides.

With the increase in computational power between the 1990s and early 2000s, the focus in aeroacoustics predictions shifted toward hybrid methods. Some developments were made in predicting noise from relatively simple turbulent flows to understand the effects of turbulence itself in far-field propagation, as reviewed by Colonius & Lele [26]. Consequently, applied research has approached such hybrid methods to decouple the noise generation problem from its propagation [27, 28, 29, 12, 30], while new formulations were being developed at the time [31, 10, 32]. This approach became popular as the unsteady near-field aerodynamics interactions were resolved using CFD without directly resolving the expensive far-field sound propagation. Aerodynamic data is extracted from CFD and fed the FW-H code, propagating the waves into the far-field observer.

More recent progress within the past 15 years has mostly focused on improving the aerodynamic step in this approach. While the research in the early 2000s tended to rely on Eulerian approaches for CFD prediction, the field has since evolved in evolved to resolving the flow fields using higher fidelity methods such as Unsteady Reynolds-Averaged Navier-Stokes (URANS) method [19, 5, 6, 33] and more recently Detached Eddy Simulations (DES) [20, 16]. With these higher fidelity methods, the noise prediction could account for noise generation from other sources such as installation effects and turbulence, which leads to better noise predictions. These advancements have also been implemented to optimise the open rotor's performance [34], while more sophisticated methods, such as the Harmonic Balance method [35] have been developed.

Despite these ongoing advancements in the CFD side of computational aeroacoustics, the permeable FW-H equation remains the foundational tool for noise prediction for aeroengine noise prediction. In this work, the FW-H method is developed and validated against fundamental sound sources. Then, the FW-H method is verified in its capability to predict higher harmonics and high-speed sources. These steps remain an important prerequisite before implementing the FW-H method into a hybrid method with CFD.

1.5 Research Objectives and Outlines

The objectives of this work are:

1. **To develop an acoustics prediction method using the permeable surface form of the FW-H equation, then validating it using fundamental sound sources**

   This part of the work is carried out in Chapter 2. The background of the permeable surface form of the Ffowcs Williams-Hawkings method is outlined, and the definition of the analytical thickness and loading (monopole and dipole, respectively) sound sources are also provided. These analytical sound sources are used as a baseline to compare the numerical FW-H results against, to build confidence in the code. The validation of the numerical FW-H method is performed using test cases that are described within the chapter.

2. **To develop an understanding of the sensitivity of the FW-H method to the temporal and spatial discretisation**

   This is carried out in Chapter 3. Using monopole sound sources, the code is further tested for its sensitivity against the temporal and spatial discretisation. Discretisation parameters are introduced early in the chapter to describe their definitions and how they are used to parametrise the different tests. There are two main tests that are conducted; i.e., mimicking the sound from a known propeller using a ring of multiple stationary monopoles (also known as a multipole); and reproducing a rotating analytical sound source with varying Mach numbers and frequency ratios using a singular rotating monopole. These tests are devised to discover the code's sensitivity to the discretisation resolutions with multiple harmonics as well as Doppler-shifted wave properties at different Mach numbers.

3. **To apply the FW-H method to predict open rotor noise using an appropriate open rotor aerodynamics CFD data**

   This is done in Chapter 4. The open rotor geometry and operating conditions are initially described to provide an understanding of the scales of the open rotor model. To understand the baseline steady performance of the engine, a steady-state CFD solution is presented to provide the integral quantities of the rotor blades such as the pressure and lift coefficients. A transient CFD case discretised based on the results from Chapter 3 is used to predict unsteady blade and rotor loadings to establish the rotor's performance in operation. Finally, the transient flow fields is used to feed the FW-H code to predict the noise radiated and propagated from the open rotor.

Chapter 2

Development and Validation of the FW-H Method

This chapter describes the details of the FW-H method and the analytical frameworks used in its validation. The formulation used in the numerical FW-H method is described in Section 2.1 below along with the description of how the computer code works. The following sections will then describe the test cases used to validate the code; describe the analytical solutions of the monopole and dipole used to compare against the numerical solution; and finally, the validation data is presented to build confidence in the numerical code itself.

2.1 Description of the FW-H Method

2.1.1 Derivation of the FW-H Equation

To predict the open rotor noise, the FW-H equation in its permeable surface form provides a useful basis for noise prediction. This formulation is based on the governing laws of fluid motion, and the descriptions provided below are based on the original paper by Ffowcs Williams and Hawkings [9]. Consider a control surface *S*, which contains a combination of fluid and solid boundaries, while the region outside contains only fluid. The control surface *S* moves at a velocity v and is defined by the equation $f(x,t) = 0$, the following relationship then hold,

$$\begin{gathered} f < 0, H(f) = 0 \\ f > 0, H(f) = 1 \\ \nabla_x f = |\nabla_x f|\mathbf{n}, \mathbf{n} \text{ is the outward normal to } S \\ D_v f/Dt = 0, \text{ so } \partial f/\partial t = -v_{\mathbf{n}}|\nabla_x f| \\ \partial H(f)/\partial x_i = \delta(f)\partial f/\partial x_i = \delta(f)\mathbf{n}_i|\nabla_x f| \end{gathered} \tag{2.1}$$

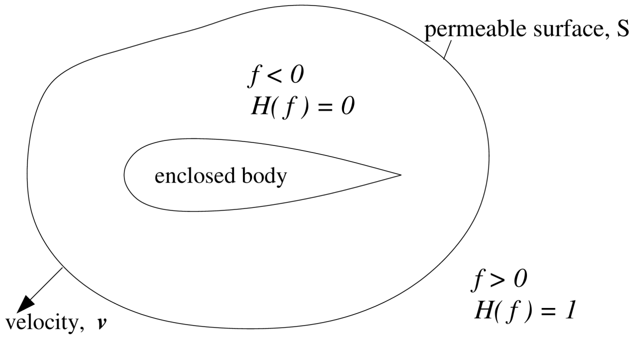


Figure 2.1: Permeable control surface

Generalised flow variables are denoted by an overbar. Outside of the surface, the variables are equal to the real flow variables, while inside they are assigned the value zero, therefore, there is a discontinuity across the surface *S*. Manipulating the mass and momentum conservation equations,

$$\begin{gathered} H(f)\left(\frac{\partial \rho'}{\partial t} + \frac{\partial \rho u_i}{\partial x_i}\right) = 0 \\ \frac{\partial \overline{\rho'}}{\partial t} + \frac{\partial (\overline{\rho u_i})}{\partial x_i} = (\rho_0 u_{\mathbf{n}} + \rho'(u_{\mathbf{n}} - v_{\mathbf{n}}))\delta(f)|\nabla_x f| \end{gathered} \tag{2.2}$$

$$\begin{gathered} H(f)\left(\frac{\partial (\rho u_i)}{\partial t} + \frac{\partial}{\partial x_j}(p_{ij} + \rho u_i u_j)\right) = 0 \\ \frac{\partial (\overline{\rho u_i})}{\partial t} + \frac{\partial}{\partial x_j}(\overline{p_{ij}} + \overline{\rho u_i u_j}) = (p_{ij}\mathbf{n}_j + \rho u_i(u_{\mathbf{n}} - v_{\mathbf{n}}))\delta(f)|\nabla_x f| \end{gathered} \tag{2.3}$$

subtracting the divergence 2.3 from the time derivative 2.2 gives an inhomogeneous wave equation,

$$\begin{aligned} \Box^2 p'(x,t) = {} & \frac{\partial^2 \overline{T_{ij}}}{\partial x_i \partial x_j} - \frac{\partial}{\partial x_i}\left(p_{ij}\mathbf{n}_j + \rho u_i(u_{\mathbf{n}} - v_{\mathbf{n}})\right)\delta(f)|\nabla_x f| \\ & + \frac{\partial}{\partial t}\left(\rho_0 u_{\mathbf{n}} + \rho'(u_{\mathbf{n}} - v_{\mathbf{n}})\right)\delta(f)|\nabla_x f| \end{aligned} \tag{2.4}$$

Which is the differential form of the FW-H equation, and is the governing equation for the generation and propagation of sound. Converting this differential form into an equivalent integral form is done using Green's function (see Appendix C). A coordinate change from the fixed one to a moving one with the control surface η is performed so the sources are at rest in the η space. The Jacobian for the coordinate change is J which represents the ratio of volume elements in the η and y spaces. Performing the steps above gives the inhomogeneous wave equation in the integral form of the FW-H equation,

$$
\begin{aligned}
p'(\boldsymbol{x},t) = {} & \frac{\partial^2}{\partial x_i \partial x_j} \int_{-\infty}^{\infty} \frac{\overline{T_{ij}}\delta(g)J}{4\pi r} d^3\eta d\tau \\
& - \frac{\partial}{\partial x_i} \int_{-\infty}^{\infty} \frac{\left(p_{ij}\mathbf{n}_j + \rho u_i(u_{\mathbf{n}} - v_{\mathbf{n}})\right)\delta(f)\delta(g)|\nabla_x f|J}{4\pi r} d^3\eta d\tau \\
& + \frac{\partial}{\partial t} \int_{-\infty}^{\infty} \frac{\left(\rho_0 u_{\mathbf{n}} + \rho'(u_{\mathbf{n}} - v_{\mathbf{n}})\right)\delta(f)\delta(g)|\nabla_x f|J}{4\pi r} d^3\eta d\tau
\end{aligned}
\tag{2.5}
$$

The first right-hand side term, the quadrupole, is not dealt with explicitly [25, 3] as it is more significant for transonic and supersonic Mach numbers [9, 22]. By choosing a sufficiently large permeable surface, their contributions outside of this surface are negligible,

$$
\begin{aligned}
p'(\boldsymbol{x},t) = {} & -\frac{\partial}{\partial x_i} \int_{-\infty}^{\infty} \frac{\left(p_{ij}\mathbf{n}_j + \rho u_i(u_{\mathbf{n}} - v_{\mathbf{n}})\right)\delta(f)\delta(g)|\nabla_x f|J}{4\pi r} d^3\eta d\tau \\
& + \frac{\partial}{\partial t} \int_{-\infty}^{\infty} \frac{\left(\rho_0 u_{\mathbf{n}} + \rho'(u_{\mathbf{n}} - v_{\mathbf{n}})\right)\delta(f)\delta(g)|\nabla_x f|J}{4\pi r} d^3\eta d\tau
\end{aligned}
\tag{2.6}
$$

Using the notation by di Francescantonio [31] to simplify the algebra, new variables U_i and L_i are introduced to represent mass and momentum fluxes through the control surface,

$$
U_i = (1 - \frac{\rho}{\rho_0})v_i + \frac{\rho}{\rho_0}u_i \quad L_i = p_{ij}\mathbf{n}_j + pu_i(u_{\mathbf{n}} - v_{\mathbf{n}})
$$

Considering an undistorted control surface that translates at a constant and subsonic velocity, as described by Brentner and Farassat [36] is,

$$
\begin{aligned}
p'(\boldsymbol{x},t) = {} & \frac{1}{4\pi} \int_S \left[\frac{\rho_0(\dot{U}_{\mathbf{n}} + U_{\hat{\mathbf{n}}})}{r(1 - M_r)^2}\right]_{\tau^*} dS(\eta) \\
& + \frac{1}{4\pi} \int_S \left[\frac{\rho_0(r(dM/d\tau)_r + c(M_r - |M|^2))}{r^2(1 - M_r)^3}\right]_{\tau^*} dS(\eta) \\
& + \frac{1}{4\pi c} \int_S \left[\frac{\dot{L}_r}{r(1 - M_r)^2}\right]_{\tau^*} dS(\eta) \\
& + \frac{1}{4\pi} \int_S \left[\frac{L_r - L_m}{r^2(1 - M_r)^2}\right]_{\tau^*} dS(\eta) \\
& + \frac{1}{4\pi c} \int_S \left[\frac{L_r(r(dM/d\tau)_r + c(M_r - |M|^2))}{r^2(1 - M_r)^3}\right]_{\tau^*} dS(\eta)
\end{aligned}
\tag{2.7}
$$

which is the numerical integration that is used by the numerical FW-H code to resolve the sound pressure at the observer locations. The first two terms describe the sound by monopole sources and the last three terms describe the sound by dipole sources.

2.1.2 Description of the FW-H Code

The FW-H code was written in Fortran 90 to implement the calculations computationally using equation 2.7. The program is summarised in the figure below,

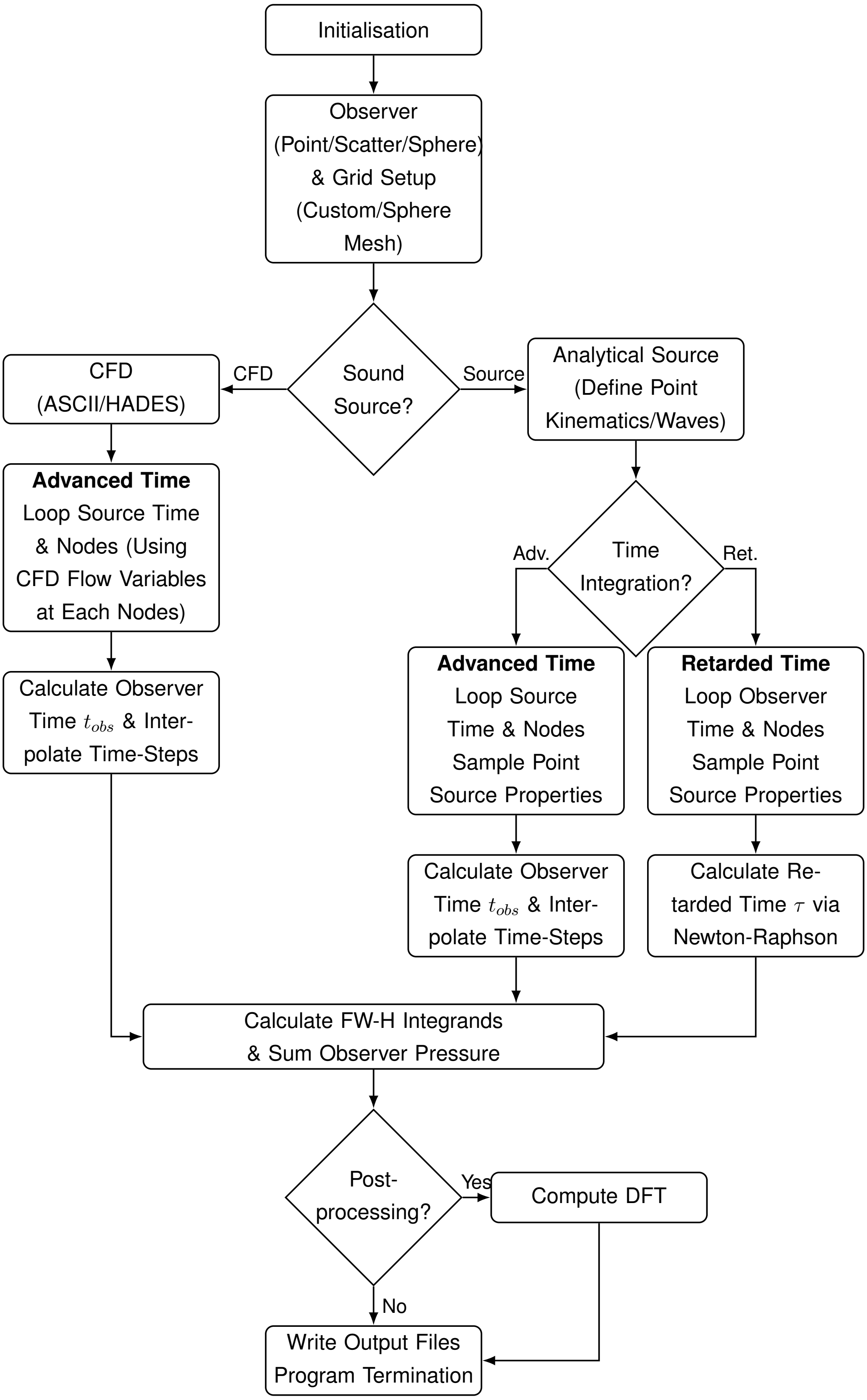


Figure 2.2: FW-H code flowchart: the code can receive both CFD data and analytical point sources as input, while using either advanced time or retarded time integration approaches

The ambient flow properties are first defined to be used as reference values. The code then checks whether the input sound source is an analytical point source, (simply referred to as, ”Source” in the code) or discrete CFD input data (referred to as ”ASCII/HADES”). The analytical point source will be described in the following sections and is used for the validation and sensitivity analyses conducted in this chapter and Chapter 3 as a reference value against which the numerical results are compared. CFD data is used in the Chapter 4 following the validation and sensitivity analysis of the FW-H code.

Following the source definition, the time integration approach is defined. Two time integration methodologies are implemented:

1. **The Retarded Time Method:** This approach is only available for the point sources. It iterates over a uniform grid of the observer's reception time, t, and implicitly calculates the corresponding source emission time τ. Because the distance between the source and observer, $r(\tau)$, is a function of an unknown emission time, the retarded time, $\tau = t - r(\tau)/c$ is resolved iteratively using a Newton-Raphson root-finding algorithm [37].
2. **The Advanced Time Method:** Also known as source-dominant formulation [32], this approach is compatible with both point sources and CFD inputs. As τ is known in this method, it instead iterates over the source's discrete emission times and computes the reception time at the observer $t = \tau + r(\tau)/c$. Because the motion of the sources may be non-uniform, the reception times are interpolated onto a uniform observer time grid.

Following the time integration, the surface integration in Equation 2.7 is resolved on the permeable surface grids to predict the observer acoustic pressure at time τ using a Riemann summation. This integration is performed at every time step to resolve the acoustic pressure time history at the observer location. Finally, the code can optionally post-process the time history data using a Discrete Fourier Transform (DFT).

It should be noted that as the surface integration and the time-marching use a finite number of discrete grids and time steps, respectively, both the validation and sensitivity studies will also evaluate how varying the spatial and temporal resolutions affects the numerical solution's accuracy. These are done to understand the code's behaviour in replicating the analytical solution, and therefore, to understand how to use it on a real CFD case.

When using the analytical source as input, the code also computes a direct analytical calculation, the descriptions of which are provided in Sections 2.3 and 2.5. These are calculated to give an analytical baseline against which the numerical FW-H results are compared. The following subchapters will discuss the validation of the FW-H code by comparing it against the known analytical solution.

2.2 Validation of the FW-H Code

To validate the numerical integration of the FW-H code, several test cases are developed where an analytical point source are used as an input. This section describes the validation methodology and the test cases. Figure 2.3 below illustrates the validation framework.

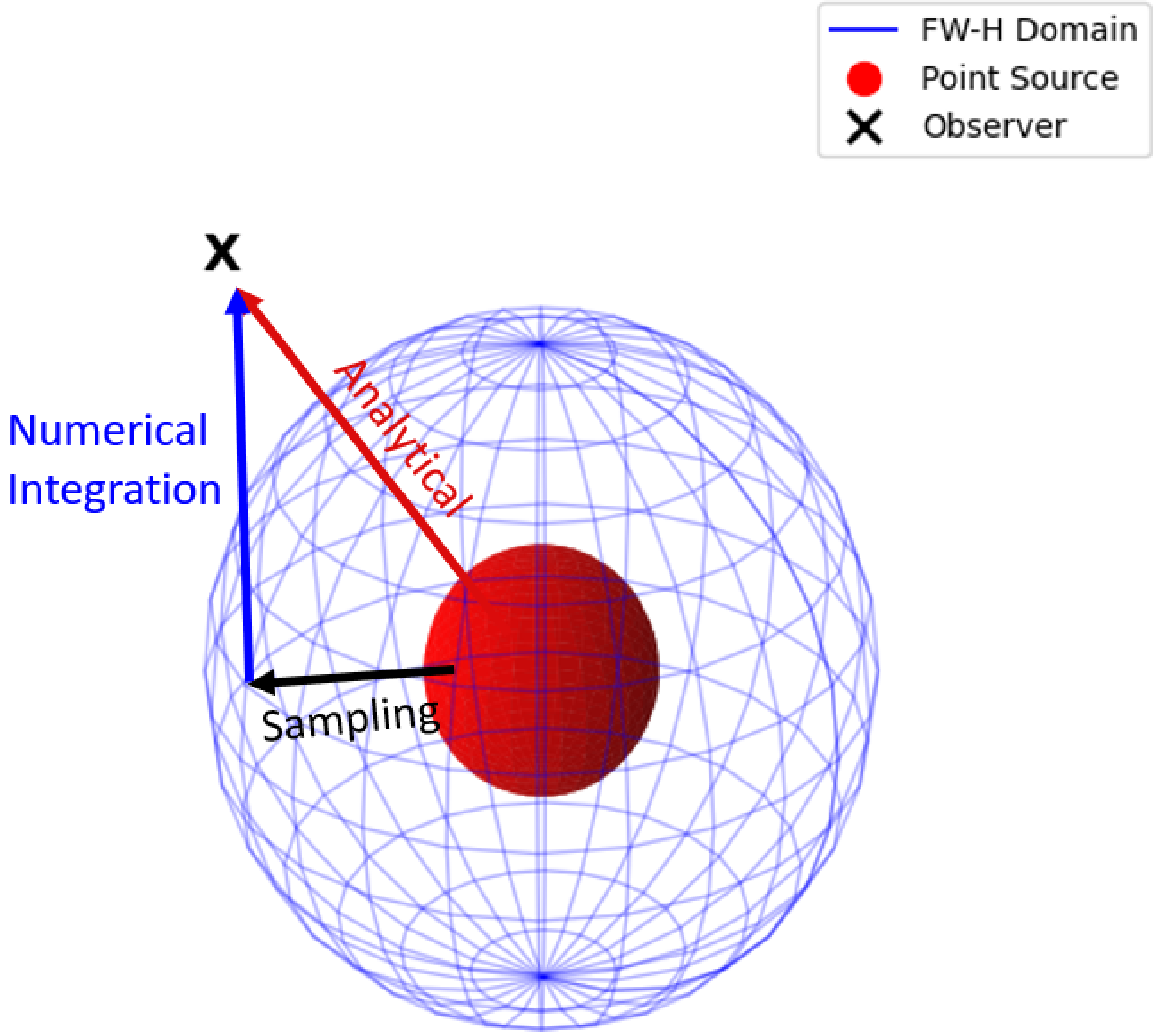


Figure 2.3: Validation framework

The numerical code samples the pressure generated by the source on the permeable boundary (black arrow), followed by surface integration of pressure to calculate the acoustic pressure at the observer (blue arrow). The analytical solution is calculated directly from the source to the observer (red arrow).

2.2.1 Test Cases

The following are the test cases used in the validation:

- Stationary and co-translating point source and observer. This test case is used as a baseline to test the overall prediction of the zero relative motion between point source and observer. The tests will also demonstrate the similarity between the two cases.

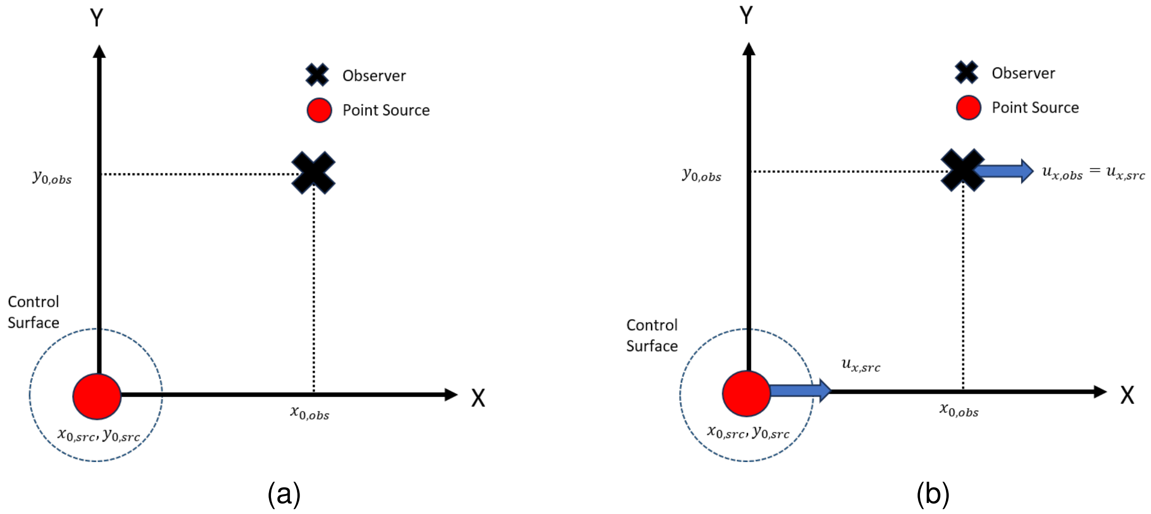


Figure 2.4: Illustration of, a) stationary point source and observer, and b) co-translating point source and observer cases

- Translating point source and stationary offset observer. This test case demonstrates the prediction with cases of non-zero relative motion between point source and observer. The observer is placed offset from the source's axis of motion such that the source can approach and move away from the observer in a single test case. This test case is done only for the monopole.

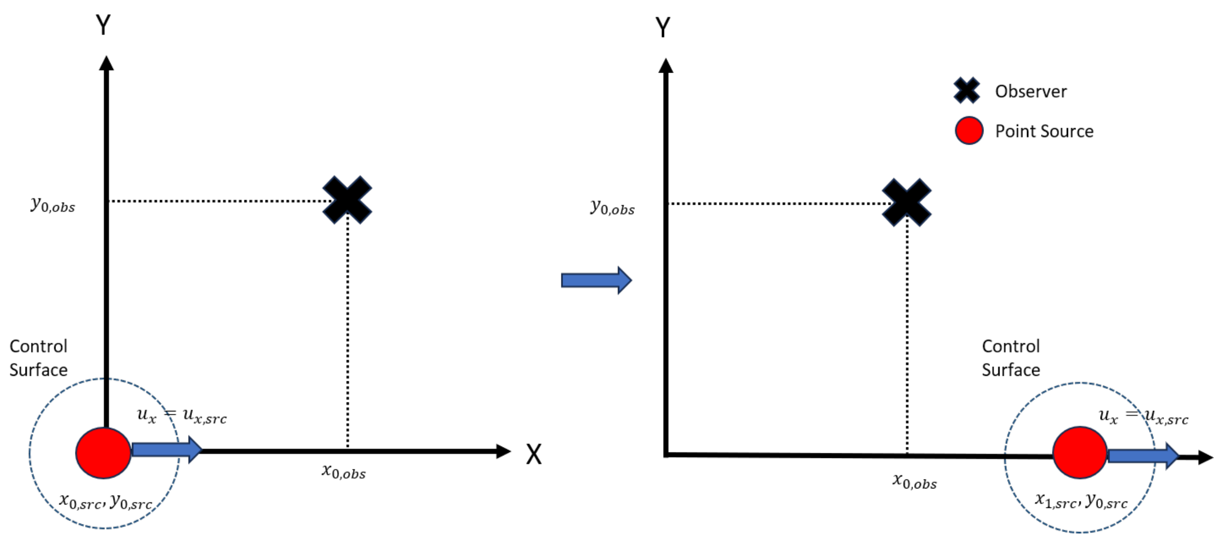


Figure 2.5: Illustration of translating point source and stationary observer cases

- Translating point source and stationary inline observer. This test case is done for the dipole test case to demonstrate the non-zero relative motion. The observer is placed inline instead of offset unlike the previous case to simplify the problem for the dipole's derivation in 1-D. The simulations are defined to finish before the control surface touches the observer to avoid singularity.

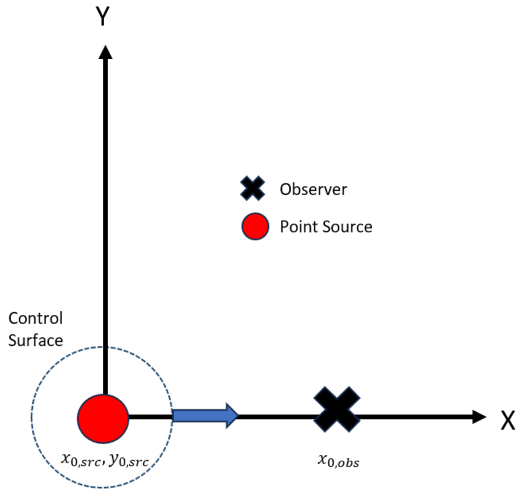


Figure 2.6: Illustration of translating point source and inline stationary observer case

2.2.2 Error Quantification

The errors of the numerical integration are quantified using the L2-norm error,

$$Error = \frac{\Sigma_j (p_{j,ref} - p_{j,num})^2}{\Sigma_j (p_{j,ref})^2} \tag{2.8}$$

which uses the same approach as Casalino's paper on the Advanced Time approach [32]. This approach is used to compare the instantaneous values of the numerical FW-H integration against the analytical model at every point in time; therefore, any deviations from the analytical results are evaluated. If the number of data points (i.e., number of time-steps) between the analytical and numerical solutions differ, such as during tests of varying temporal resolutions, the coarser numerical results are linearly interpolated to make up for the missing time-steps.

2.3 Analytical Solution for a Monopole

Analytical approaches for monopole sources have been validated in previous work by Char and Thomas [37, 18]. This section describes the derivation of the analytical monopole to provide a background on the analytical solution provided in the validation cases. Monopoles or thickness sources are sounds that are equivalent to mass supply, described as the following,

$$\Box^2 p'(\boldsymbol{x},t) = \frac{\partial}{\partial t}(Q(t)\delta(\boldsymbol{x} - y(t))) \tag{2.9}$$

Convolving the above with the Green's function in 3D free-space (see Appendix C, described as,

$$G = \frac{\delta(g(\tau))}{4\pi r} \tag{2.10}$$

where $g(\tau) = t - \tau - \frac{r(\tau)}{c}$ and $r(\tau) = |\boldsymbol{x} - y(\tau)|$, gives the following,

$$p'(\boldsymbol{x},t) = \frac{\partial}{\partial t}\left[\frac{Q(\tau)}{4\pi r(1-M_r)}\right]_{\tau*} \tag{2.11}$$

which is the description of the monopole used in this work.

2.3.1 Monopole Pressure Equation

Taking the monopole definition as described in Equation 2.11, taking the differentiation using the definitions as described in Appendix B, the analytical solution for pressure is obtained,

$$p'(\boldsymbol{x},t) = \frac{1}{4\pi}\left[\frac{\partial Q(\tau)/\partial\tau}{r(1-M_r)^2} + \left(\frac{\partial M}{\partial \tau}\right)_r \frac{Q(\tau)}{r(1-M_r)^3} + \frac{Q(\tau)c(M_r - |M|^2)}{r^2(1-M_r)^3}\right]_{\tau*} \tag{2.12}$$

The first term on the right-hand side describes the far-field propagation of the monopole due to its strength's temporal gradient, $\partial Q/\partial\tau$, the second term describes the far-field propagation due to acceleration, $\partial M/\partial\tau$, and the third term describes the near-field propagation.

2.3.2 Monopole Velocity Equation

Using the momentum equation,

$$\rho_0\frac{\partial u_i}{\partial t} + \rho_0\left(u_i\frac{\partial u_i}{\partial x_j}\right) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left(\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right) \tag{2.13}$$

Linearising the equation and assuming no mean flow, substituting the monopole definition into the pressure term, and convolving with Green's function to replace $\partial/\partial x_i$ terms with

$\partial/\partial t$ (see Appendix D), the solution for velocity is obtained,

$$
\begin{aligned}
u_i' = \frac{1}{4\pi\rho_0}\Bigg[&\frac{(\partial Q(\tau)/\partial\tau)r_i}{cr^2(1-M_r)^2} + \frac{Q(\tau)r_i}{r^3(1-M_r)^3} + \left(\frac{dM}{d\tau}\right)\frac{Q(\tau)r_i}{cr^2(1-M_r)^3} \\
&- \frac{Q(\tau)M^2 r_i}{r^3(1-M_r)^3} - \frac{Q(\tau)M_i}{r^2(1-M_r)^2}\Bigg]_{\tau*}
\end{aligned}
\tag{2.14}
$$

where the first and third terms correspond to the far-field velocity perturbation components due to the strength and Mach number temporal gradients, respectively; and the second, fourth, and fifth terms correspond to the near-field perturbation components.

2.3.3 Monopole Density Equation

The expression for density is based on the assumption of a linear, isentropic flow. Therefore, the expression for density is,

$$\rho' = \frac{p'}{c^2} \tag{2.15}$$

which is valid for both monopole and dipole sources.

2.4 Validation of the FW-H Method for Monopole Sources

The point source in each case is prescribed with source strength of $Q(\tau) = \sin(10\tau)$, and is placed at the origin. Speed of sound, c, and the freestream air density ρ_0 are set to $340m/s$ and $1.2kg/m^3$, respectively. The Retarded and Advanced Time algorithms are both tested.

Three cases are defined to validate the monopole; the co -stationary, co-translating, and the relative motion cases. These cases provide an understanding of the code's ability in predicting simple sounds. The observer is initially located at $[10, 10, 0]$ for the co-stationary and co-translating cases, and at $[150, 10, 0]$ for the relative motion case. In the translating cases, the relevant points are moving at Mach 0.5 along the x-direction. The test cases are summarised in Table 2.1 below.

Table 2.1: Summary of Monopole Validation Test Cases

Case Name	**Time Algorithm**	M_{source}	$M_{observer}$	$x_{0,observer}$
Co-Stationary	Retarded & Advanced	[0, 0, 0]	[0, 0, 0]	[10, 10, 0]
Co-Translating	Retarded	[0.5, 0, 0]	[0.5, 0, 0]	[10, 10, 0]
Translating	Retarded	[0.5, 0, 0]	[0, 0, 0]	[150, 10, 0]
Translating	Advanced	[0, 0, 0]	[0.5, 0, 0]	[150, 10, 0]

The radius of the control surface is set to $5m$ and the surface is discretised to capture the pressure wavelengths at 275, 100, and 35 grid points per wavelength, which correspond to 40×40, 15×15, and 5×5 grid resolutions, respectively. Similarly, the temporal discretisation is set to capture frequencies at 30, 15, and 5 time steps per wave period.

2.4.1 Co-Stationary Monopole Case

Starting with the co-stationary case using the Retarded Time algorithm, it can be seen from Figure 2.7 below that the numerical FW-H method agrees well with the analytical solution; provided that the discretisation resolutions resolve the pressure waves radiated by the point source sufficiently. An under-prediction of the amplitudes are observed when using coarser grid resolutions, while using coarser temporal resolutions causes the FW-H prediction to lag behind the analytical solutions.

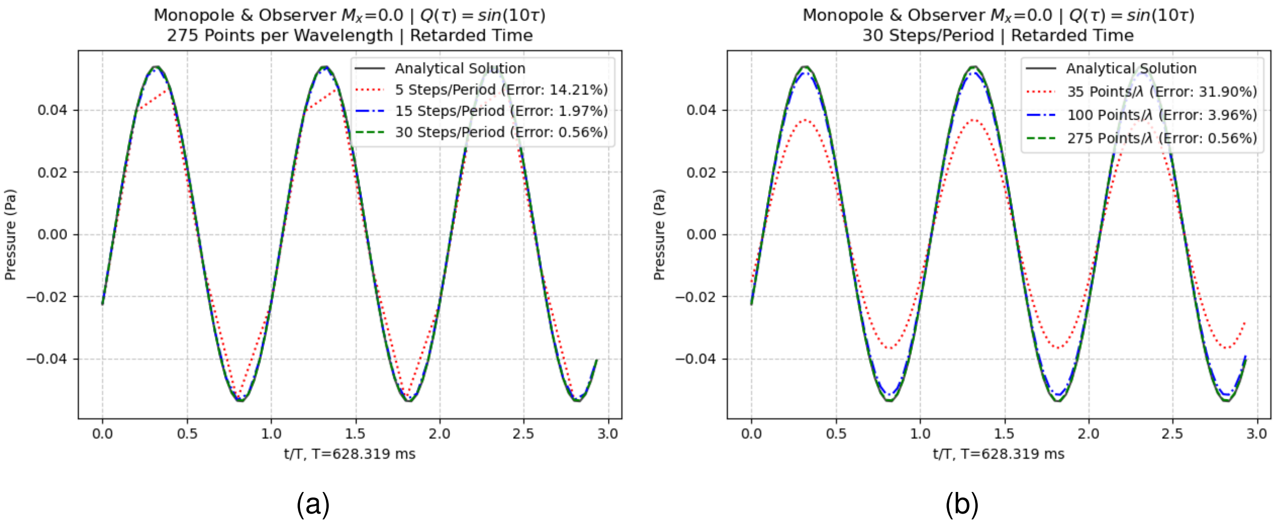


Figure 2.7: Co-stationary monopole case using the retarded time algorithm, varying the a) time-step sizing, and b) grid sizing

Regarding the amplitude under-prediction, it cannot be attributed to the code's inability to predict the wavelengths; as the wavelength in this case is ~ 213.6 meters, while the coarsest 5×5 grid resolution resolves this to 35 points per wavelength. Instead, this is largely a geometric error caused by the coarser grids that is approximating the control surface sphere as a polygon. On the other hand, the coarsest temporal resolution in this case was indeed unable to predict the waves well enough due to the insufficient temporal resolution.

Moving on to the Advanced Time algorithm, it should be noted that due to the nature of this algorithm, the sound signal received at the observer requires some post-processing work. This post-processing removes the first and last period of the observer signal which reads as zero. This post-processing was done prior to any error calculations or plotting; and therefore, all the results presented here have been cleaned up.

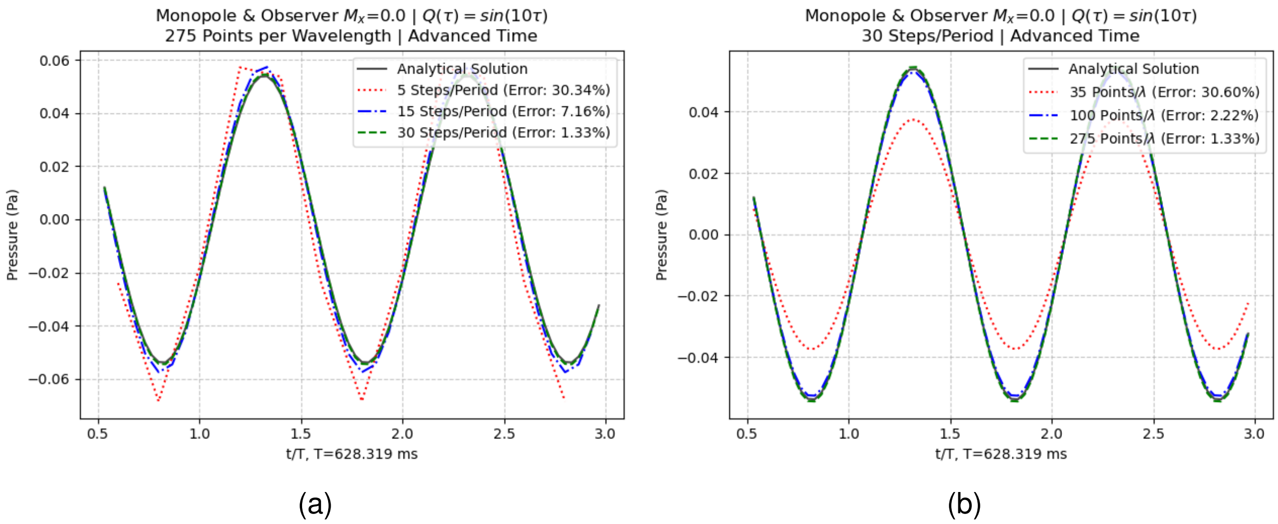


Figure 2.8: Co-stationary monopole case using the advanced time algorithm, varying the a) time-step sizing, and b) grid sizing

As with the Retarded Time results, the numerical integral using the Advanced Time method generally has a good agreement with the analytical solution. The errors are relatively higher with this method due to the time interpolation at the observer points; an inherent feature of the Advanced Time approach as described in Subsection 2.1.2. The effects of discretisation resolutions are also relatively similar to the Retarded Time solutions, where coarser time-step sizes somewhat lags behind the analytical solution, while the coarser grids under-predict the amplitude due to geometric error.

2.4.2 Co-Translating Monopole Case

The co-translating case only uses the Retarded Time algorithm as its purpose is to demonstrate how the signal is received at the observer when compared to the co-stationary case.

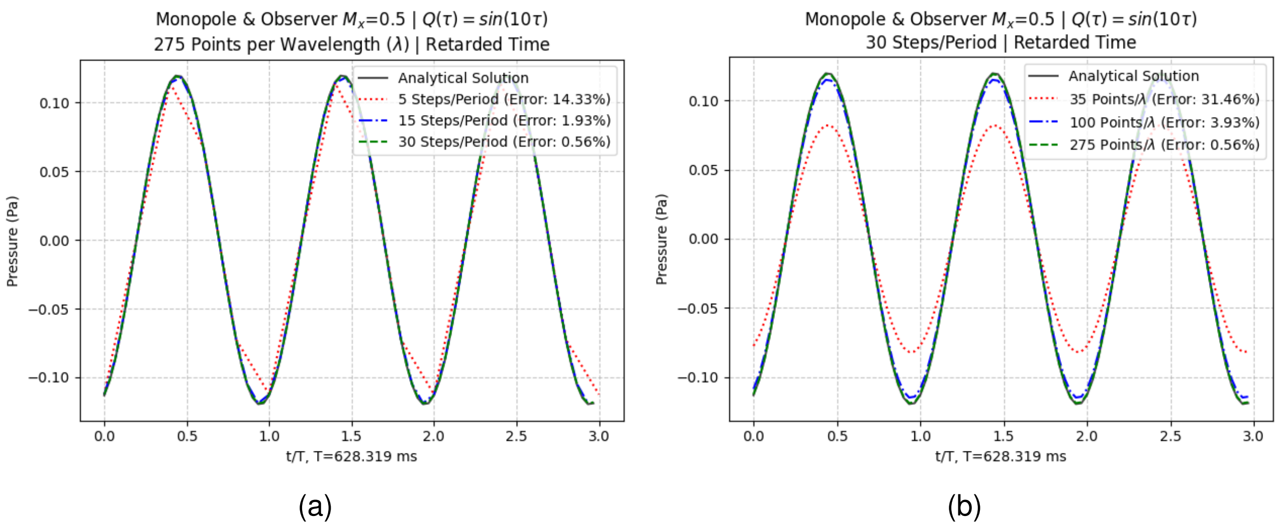


Figure 2.9: Co-translating monopole case using the retarded time algorithm, varying the a) time-step sizing, and b) grid sizing

Comparing these results with the co-stationary cases (Figure 2.7) shows a distinct phase difference and amplitude, despite both scenario having zero relative Mach number and the distance between the source and observer is kept the same. In the co-translating case, the source moves through the quiescent fluid medium, which causes a phase shift of $\sim \pi/2$ and an amplitude increase due to the convective term. Furthermore, the emission distance r in the retarded time, $\tau = t - r/c$, differs from the geometric distance because both points are moving through the medium during the propagation time. In terms of discretisation resolutions, the errors between this co-translating case and the retarded time co-stationary case are roughly the same, which suggest that despite there being some Doppler shifting due to the motion of the source, the observer's equal motion cancels out the errors which may arise.

2.4.3 Translating Monopole Case

For the translation case, the set up for the two time algorithms are slightly different, as summarised in Table 2.1. In the Retarded Time case, the source moves along the X-axis to the positive direction, while the observer remains at its initial position; while in the Advanced Time case, the source remains stationary while the observer moves parallel to the X-axis to the negative direction.

One reason for this differentiated approach is, at the time of the writing of this work, the Advanced Time algorithm does not support the permeable surface's tracking of the source's motion. This means, while the source can move within the permeable surface, the surface can only remain stationary in space; which is also the reason why the co-translating case was only tested using the Retarded Time algorithm. Another reason for this differentiated approach is to demonstrate the signal amplification due to Doppler shifts, which will be elaborated in the Advanced Time results.

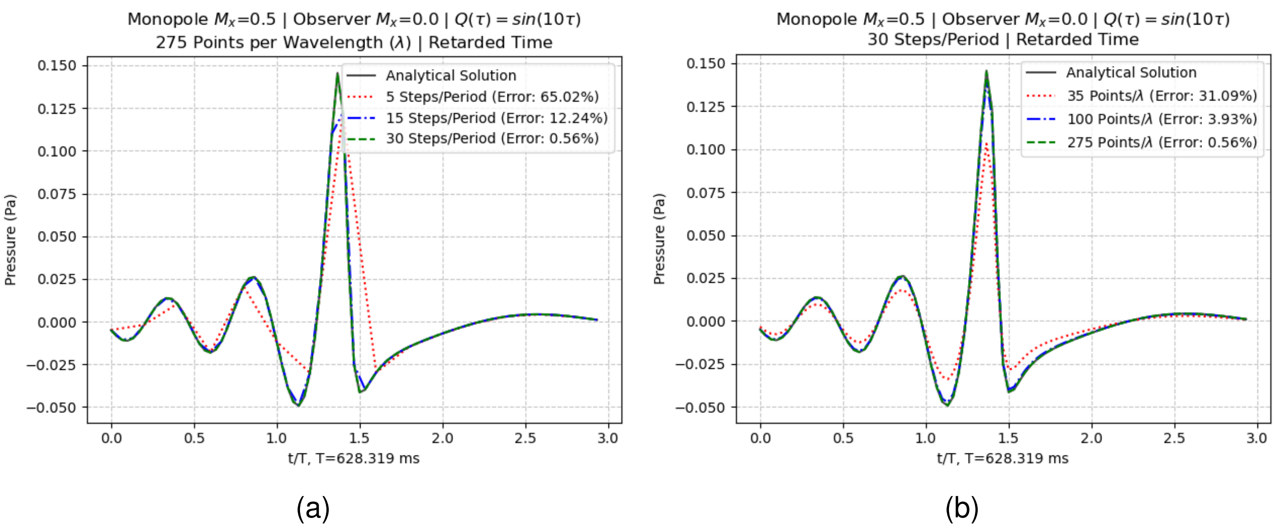
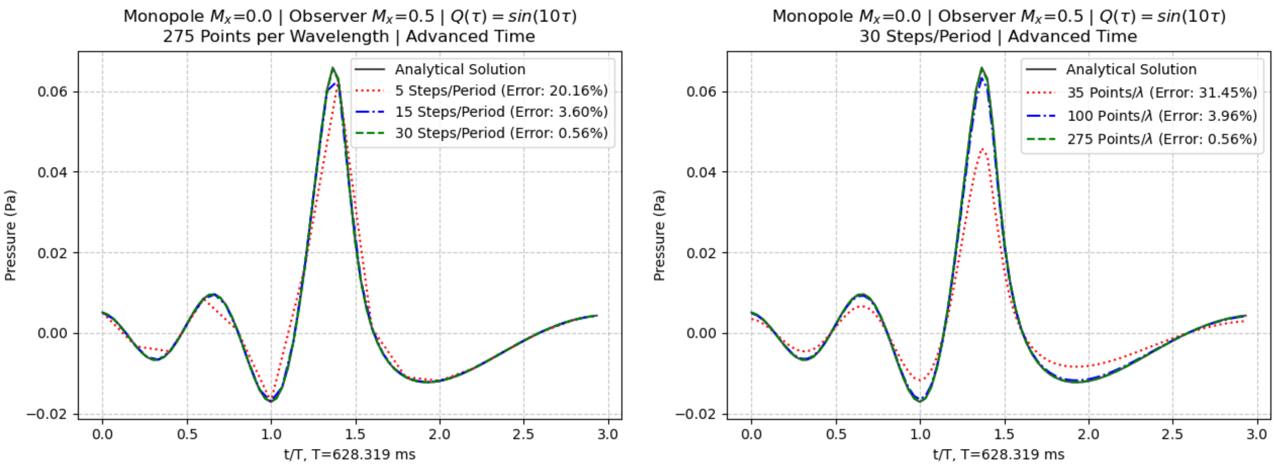


Figure 2.10: Translating monopole with stationary observer test case using the Retarded Time method, a) with varied time-step sizing, and b) with varied grid sizing

This case demonstrates the relative motion between the source and observer by imposing a relative Mach number between the two points. In Figure 2.10 the Doppler effect causes the perceived frequency and pressure amplitude at the observer to increase as the source approaches ($0 \leq t/T \leq 1.5$), and decrease as it moves away ($t/T > 1.5$). This wave compression and expansion effect of Doppler shift is a common occurrence in aeroengines, whereby as the blade rotates, its motion causes the blade to approach and move away from a stationary observer. As the Figure 2.10 above shows, the Retarded Time algorithm can predict this Doppler shift effects with sufficient discretisation resolutions.

Moving to the Advanced Time algorithm, when compared to the Retarded Time algorithm, one noticeable feature of the time-history signal of the former is the lack of the large amplitude spike as the observer moves towards the source.



(a) (b)

Figure 2.11: Translating observer with stationary monopole test case using the Advanced Time method, a) with varied time-step sizing, and b) with varied grid sizing

As can be seen from Figure 2.11, the frequency shifts as the observer is moving, but the pressure amplification peak only reaches $\sim 0.07Pa$, instead of $\sim 0.150Pa$ as in the Retarded Time case. This is not an issue with the Advanced Time algorithm itself, but in how the case was set up. This pressure amplification by a factor of $1/(1 - M_r)$ is most efficient in the direction of the source's motion, and gradually reduces to $1/(1 + M_r)$ in the opposing direction of the motion. However, it can be said from this result that the Advanced Time algorithm can be used to predict this Doppler shifting effect with a comparable accuracy to the Retarded time algorithm.

With these cases explored, it can be concluded that the FW-H code can sufficiently predict the sound signal coming from a thickness (monopole) noise. The next sections within this chapter will focus on the loading (dipole) noise sources, which is highly relevant the noise generated from an open rotor engine.

2.5 Analytical Solution for a Dipole

Dipoles or loading sources are sounds due to forces acting on the fluid, equivalent to body forces and momentum supply. It is also equivalent to two equal monopoles with opposite signs at an infinitesimally small distance. The dipole is described as the following,

$$p'(\boldsymbol{x},t) = -\frac{\partial}{\partial x_i}\left[\frac{f_i(\tau)}{4\pi r(1-M_r)}\right]_{\tau*} \tag{2.16}$$

In the dipole description above, the source forcing term f_i is not an arbitrary vector. To analytically model a dipole, the mathematical equivalence between monopoles and dipoles must be established. Sound pressure generated by a monopole source is defined as,

$$p'_{mono}(\boldsymbol{x},t) = \frac{\partial}{\partial t}\left(\frac{Q(\tau)}{4\pi r|1-M_r|}\right)$$

By describing a dipole as two monopoles of equal strength and opposite signs separated by an infinitely small distance δ, the dipole pressure is a superposition of the two monopoles,

$$p'_{dipole} = p'_{mono}(\boldsymbol{x},t) - p'_{mono}(\boldsymbol{x}-\delta,t)$$

As δ tends to zero, this distance can be expressed as a first-order Taylor series expansion,

$$p'_{dipole}(\boldsymbol{x},t) \approx -\delta\frac{\partial}{\partial x_i}\frac{\partial}{\partial t}\left[\frac{Q(\tau)}{4\pi r|1-M_r|}\right]$$

Hence, the dipole source strength $f_i(t)$ can be defined as,

$$f_i(t) = 4\pi r|1-M_r|\delta\frac{\partial}{\partial t}\left(\frac{Q(t)}{4\pi r|1-M_r|}\right) \tag{2.17}$$

by defining the forcing term in this term and evaluating it using central differencing, the analytical code captures the expansion of this forcing term without requiring an analytical expansion.

2.5.1 Dipole Pressure Equation

Using the definition described in Equation 2.16, replacing the $\partial/\partial x_i$ term as outlined in Appendix D, and using the steps from monopole pressure, the pressure for an analytical dipole is obtained,

$$
\begin{aligned}
p'(\boldsymbol{x},t) = &\frac{(\partial f_i(\tau)/\partial\tau)r_i}{4\pi c r^2(1-M_r)^2} + \frac{f_i(\tau)r_i}{4\pi r^3|1-M_r|^3} - \frac{f_i(\tau)M_i}{4\pi r^2|1-M_r|^2} \\
&+ \left(\frac{\partial M}{\partial\tau}\right)_r \frac{f_i(\tau)r_i}{4\pi c r^2|1-M_r|^3} + \frac{f_i(\tau)r_i M_j^2}{4\pi r^3|1-M_r|^3}
\end{aligned}
\tag{2.18}
$$

where the first and fourth terms on the right-hand side are the far-field propagation due to changes in the source forcing term and Mach number, respectively; while the second, third, and fifth terms are the near-field propagation terms. The far-field terms are the primary reason for the noise propagation in rotors such as helicopter blades, propellers, and open rotor engines; as the blades rotate and push the air surrounding it, the local forcing strength and Mach number changes with time.

2.5.2 Dipole Velocity Equation

To analytically derive the velocity for a dipole, explicitly integrating the momentum equation like with the monopole is unviable, as this approach will return too many terms. Instead, limited solutions are obtained by applying some physical assumptions. The velocity derivations assume either a zero relative Mach number or a plane wave condition. An alternative method that uses the superposition of two equal monopoles of opposite signs will also be presented.

Dipole Velocity Equation: Stationary Assumption

Assuming inviscid and no mean flow in the momentum equation, the pressure gradient is,

$$
-\frac{\partial p}{\partial x_i} = \frac{\partial^2}{\partial x_i \partial x_j}\left[\frac{f_i(\tau)}{4\pi r(1-M_r)}\right]_{\tau *}
$$

Hence, the velocity is defined as,

$$
u_i' = \frac{1}{4\pi\rho_0}\frac{\partial^2}{\partial x_i \partial x_j}\int_{-\infty}^{t}\frac{f_i(\tau)}{r(1-M_r)}dt
$$

substituting the integrand to use the time at the observer, τ, and assuming zero relative velocity; therefore, $dr/dt = 0$ and $M_r = 0$. These assumptions are what causes this assumption to only be valid when the source is stationary; such that a co-translation with the observer will still violate this assumption.

To resolve the integral above, let $F_i(\tau) = \int f_i(\tau)d\tau \implies f_i(\tau) = \dot{F}_i(\tau)$, the analytical solution for dipole velocity that is valid for cases with zero relative Mach number is obtained,

$$u_i' = \frac{1}{4\pi\rho_0}\left(\frac{F_j(3\mathbf{n}_i\mathbf{n}_j - \delta_{ij})}{r^3} + \frac{f_j(3\mathbf{n}_i\mathbf{n}_j - \delta_{ij})}{cr^2} + \frac{\dot{f}_j(\mathbf{n}_i\mathbf{n}_j)}{c^2 r}\right) \tag{2.19}$$

where the first, second, and third terms on the right hand side are the near-, intermediate-, and far-field velocity propagation, respectively.

Dipole Velocity Equation: Plane Wave Assumption

The plane wave assumption allows relative Mach number between the source and observer, provided the control surface is in far-field that it encapsulates the source's entire wavelength. In the far-field, the pressure and velocity field perturbation components are in-phase, which is the reason of this requirement. The far-field is defined as,

$$kR \gg 1 \tag{2.20}$$

where k is the wave number. Physically, this means the acoustic wavelength must be smaller than the control surface diameter, allowing the entire wavelength to be fully captured by the permeable sphere. Using the derivations in Appendix G, the velocity equation is obtained,

$$u_i' = \frac{\mathbf{n}_i}{\rho_0 c} p'(\boldsymbol{x}, t) \tag{2.21}$$

hence showing that the velocity perturbation component is in-phase with the pressure.

Dual Monopole

An alternative approach to validate dipole sources is the dual monopole method. This approach entirely avoids the assumptions that were made in the previous derivations and instead uses two monopoles to create a dipole. This bypasses the limitations imposed by the assumptions; therefore, this method will remain accurate when used on cases that would violate the above assumptions.. However, to ensure that the dual monopole approach itself is valid, it needs to be compared against the analytical solutions above.

Using the definition of a dipole, this method uses two equal monopoles of opposing signs, separated by a small distance δ, which in this case is defined as,

$$k\delta \ll 1 \tag{2.22}$$

where k is the wave number. To simplify the validation process, it is that the distance between the two monopoles is $\delta = 0.01/k$, which still validates this method. Another advantage of this approach is that it allows the prediction of the directivity patterns of the dipole, the results of which will also be presented in the next section.

2.6 Validation of the FW-H Method for Dipole Sources

With the FW-H numerical solver validated for monopoles, the validation is extended to dipole sources. As mentioned in Section 2.5, explicitly deriving the velocity terms for a moving dipole is mathematically prohibitive. This necessitates testing the stationary and plane wave assumptions separately to identify their limitations. The dual monopole approach is then used to simulate the same cases as a physics-based baseline. Table 2.2 below summarises the test cases for the dipole.

Table 2.2: Summary of Dipole Validation Test Cases

No	**Assumption**	R	M_{source}	$M_{observer}$	$x_{0,observer}$	**Validity**
1	Stationary	$5.0m$	[0,0,0]	[0,0,0]	[500,0,0]	Valid
2		$5.0m$	[0.5,0,0]	[0,0,0]	[500,0,0]	Invalid
3	Plane Wave	$500.0m$	[0.5,0,0]	[0,0,0]	[1500,0,0]	Valid
4		$5.0m$	[0.5,0,0]	[0,0,0]	[1500,0,0]	Invalid
5	Dual Monopole	$5.0m$	[0,0,0]	[0,0,0]	[500,0,0]	-
6		$5.0m$	[0.5,0,0]	[0,0,0]	[500,0,0]	-
7		$500.0m$	[0.5,0,0]	[0,0,0]	[1500,0,0]	-
8		$5.0m$	[0.5,0,0]	[0,0,0]	[1500,0,0]	-

As with the monopole cases, the dipole sources radiate with a strength density of $Q(\tau) = sin(10\tau)$ and start at the origin. All cases use the Retarded Time algorithm, allowing the control surface to track the source (see Subsection 2.4.3). For the $5.0m$ permeable surface radius cases, the discretisation matches the finest resolution from the monopole validation (275 grid point per wavelength and 30 time-step per period). The plane wave assumption requires a much larger control surface radius to be valid ($500.0m$), so the resolution is relaxed to 10 grid points per wavelength. This coarser grid resolution remains sufficient for agreement with the analytical solution.

The dual monopole approach test the exact same test cases to those used for the stationary and plane wave assumption, as seen on Table 2.2. Consequently, the dual monopole results are presented alongside the analytical assumptions to compare the validity of this approach as a baseline. For all dual monopole cases, the two sources are separated along the x-axis with a distance $\delta = 0.01/k$ between the two points.

2.6.1 Stationary Assumption Validation

Figure 2.12a demonstrates that when the stationary assumption holds ($M_x = 0.0$), the numerical FW-H prediction matches the analytical solution very well. However, violating this assumption by introducing a translation of $M_x = 0.5$ causes the prediction to fail, which introduces phase lag and amplitude errors. This occurs because the stationary derivation omits the changes in the radiation distance term, $\partial r/\partial \tau = 0$, that the convective amplification and Doppler shifts are not correctly calculated in the solutions.

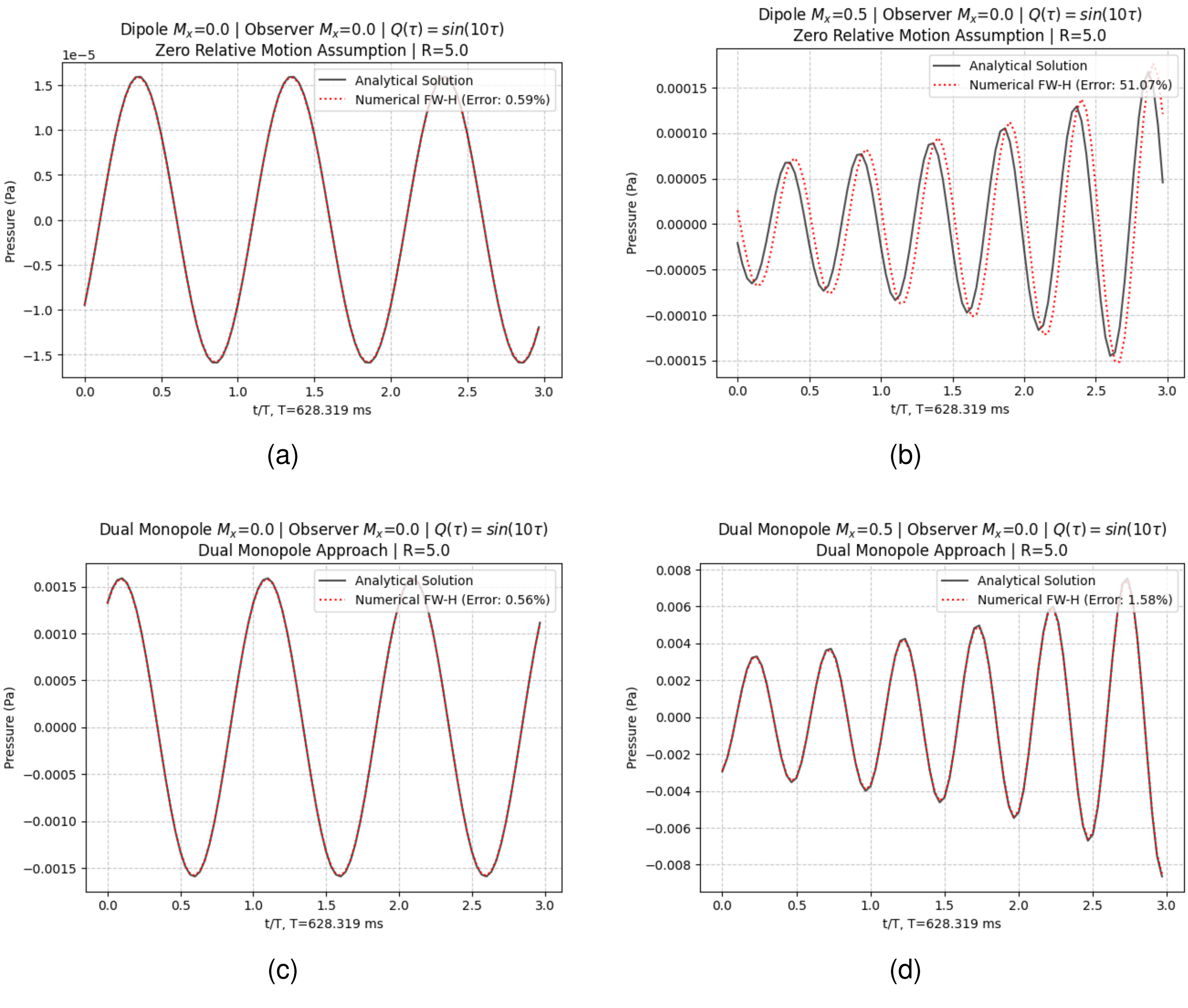


Figure 2.12: Validation of the stationary dipole assumption and the dual monopole. Top: stationary dipole assumption, with a) following, and b) violating the assumption. Bottom: dual monopole method, with c) following, and d) violating the assumption.

The dual monopole approach (Figures 2.12c and 2.12d) shows excellent agreement between its own numerical and analytical solutions regardless of source motion, as it relies only on the monopole derivations. However, comparing the dual monopole results to the pure analytical point dipole shows amplitude and phase differences. This is due to how the dipole's strength is defined in either cases; where it uses a first-order Taylor series approximation for the analytical point dipole where the distance approaches zero. Conversely, the dual monopole uses a finite physical gap δ, which introduces a phase lag of $\Delta\phi = k\delta$ between the two sources, which is not present in the analytical point dipole. Such difference between the two solutions can be remediated either using a higher-order Taylor series to define the analytical dipole's strength, or by including a phase lag definition in its approximation. Despite this geometric error, both methods have successfully validated the FW-H code's capability for a stationary dipole source.

2.6.2 Plane Wave Assumption Validation

The plane wave assumption is only mathematically valid when the signal is captured in the acoustic far-field. Therefore, the permeable surface must be sized to fulfil the $kR \gg 1$ requirement, where pressure and velocity are in-phase. As a result, the accuracy of the FW-H prediction relies on the integration surface being placed sufficiently far from the source. The results shown in Figures 2.13a and 2.13b exhibits this very well, where the numerical integration results are sensitive to the permeable surface's size relative to the source's wavelength.

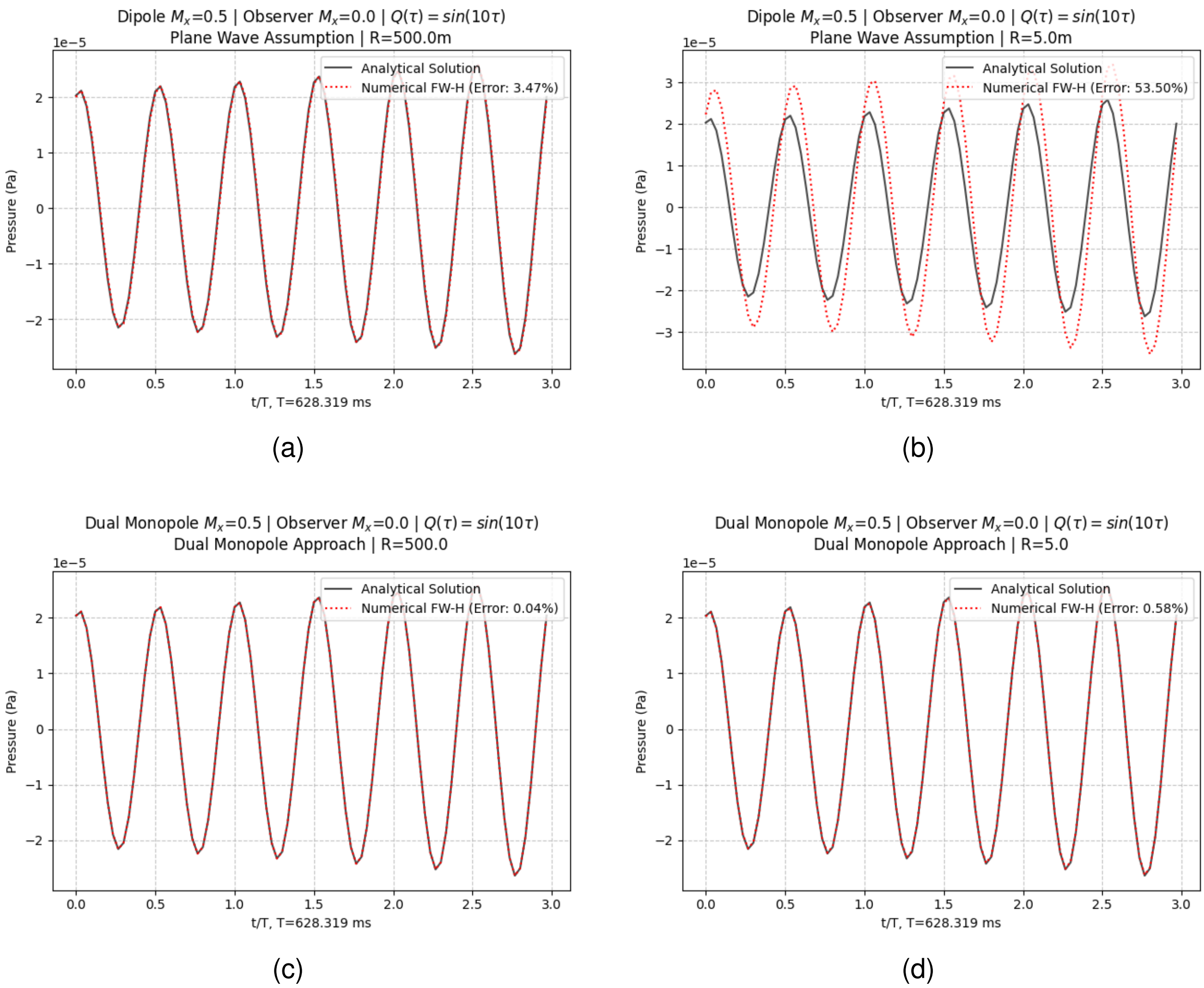


Figure 2.13: Validation of the plane wave dipole assumption and the dual monopole. Top: plane wave dipole assumption, with a) following, and b) violating the assumption. Bottom: dual monopole method, with c) following, and d) violating the assumption.

The wavelength in these test cases are $\sim 213m$, by sizing the surface radius to $500.0m$, the numerical FW-H code accurately predicts the pressure at the observer ($\sim 3.5\%$ error), even with a coarse spatial resolution (Figure 2.13a). This accuracy was received despite the coarse grid resolution at only 10 gird points per wavelength. Conversely, shrinking the control surface to $5.0m$, which violates the plane wave assumption, places the integration boundary within the near-field (Figure 2.13b). In the near-field, the pressure and velocity fields are realistically not in-phase, which causes the FW-H code to incorrectly predict the amplitude while introducing a phase-lag into the numerical prediction. Placing the integration boundary within the near-field in this case essentially forces the plane velocity prediction to be in-phase with the pressure (see Equation 2.21), which is physically incorrect.

This boundary radius sensitivity is not present in the dual monopole approach. Because this approach uses the solution for the monopole, it directly captures both the near- and far-field propagation of the pressure waves, and is independent of its pressure equation. This allows the numerical FW-H integral to sample the phase difference between the velocity and pressure fields correctly. Consequently, the dual monopole approach predicts the pressure at the observer regardless of whether the control surface radii is within in the near- or far-field of the source (Figure 2.13c and 2.13d).

An interesting behaviour is observed when comparing the analytical solutions of the point dipole and the dual monopole in these cases. Unlike the stationary validation cases where the two analytical baselines showed an amplitude and shift, the analytical solutions between each case here agree very well. Because the observer is placed significantly further away at $1500.0m$, the geometric phase lag caused by the gap between the dual monopoles become negligible. From an observer at such a far away distance, the discrete two point monopoles assembly is physically indistinguishable from a single point dipole.

The following subsection evaluates the sound directivity, which is a property of the dipole sound sources that is possible to model using the dual monopole approach.

2.6.3 Dipole Directivity using Dual Monopole Approach

The dual monopole method allows the investigation of the directivity patterns of a dipole. Sound directivity describes how the sound propagates in space due to the varying sound generation profiles at the source. This is highly relevant to the open rotor as the sound propagation from such system varies in space due the radiation profiles and kinematics.

Two standard tests are conducted, where the dipole is stationary and another that it moves at $u_x = 10m/s$ to demonstrate the convective effects. The observers are set as a sphere located at a distance of $50m$, from the origin. In either case, the sources are aligned along the global y-axis at a distance $\delta = 0.01/k$.

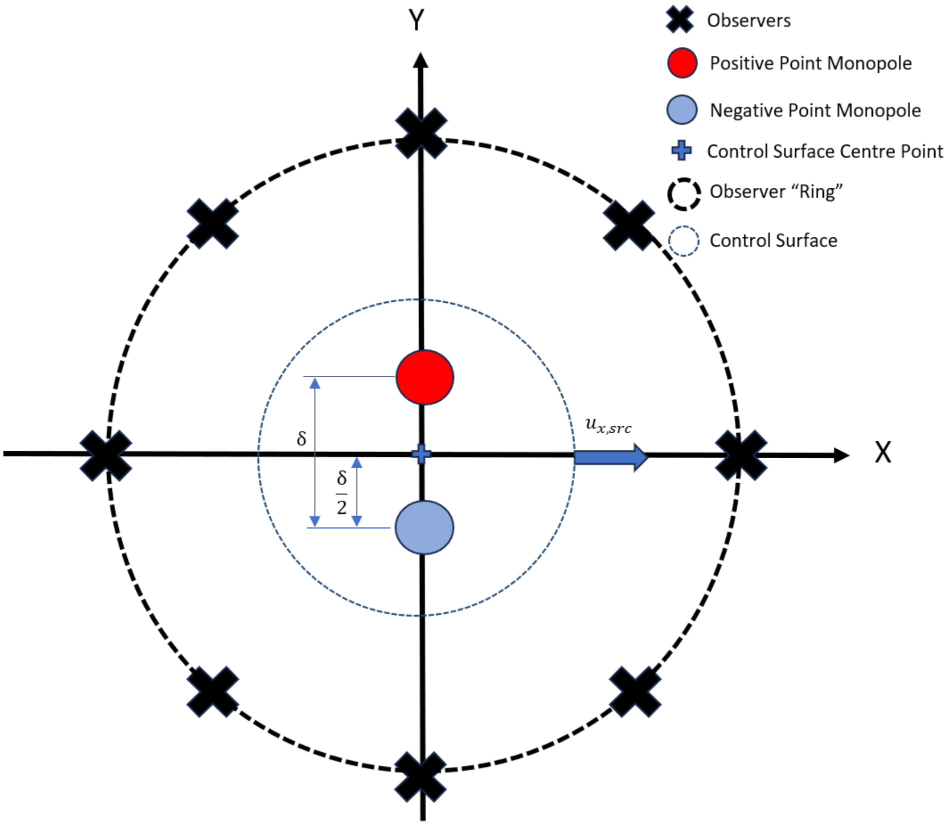


Figure 2.14: Test case for directivity

The stationary case show a standard directivity "figure-8" pattern, showing the locality of the individual monopoles. Imposing a translation to the source alters this pattern towards the x-axis as the source moves in said direction. It can be seen in either case that as the sound pressure along the x-axis is essentially 0. This is due to how the dipole itself is defined, where the two monopoles have equal and opposite signs, which causes the pressure along the x-axis to cancel.

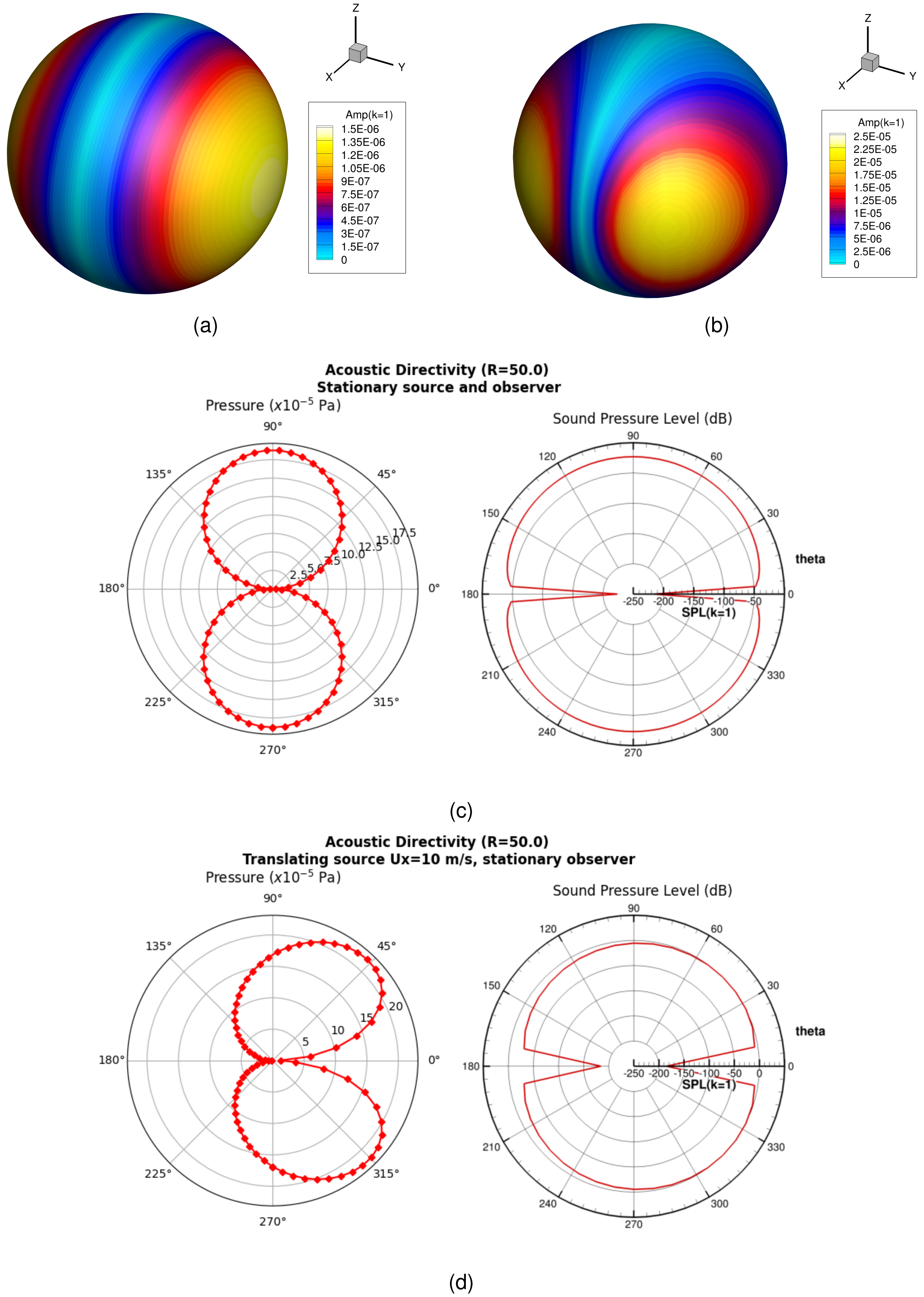


Figure 2.15: Directivity of dual monopoles, stationary case with a) surface plot and c) polar plot; and translating case with b) surface plot and d) polar plot

2.7 Concluding Summary

In this chapter, the development of the FW-H method has been described. Test cases have been developed to validate sources in stationary and motion. Both the thickness and loading noise, which are the sources that are evaluated in the permeable surface formulation, were validated whilst testing the limits of time integration algorithms, discretisation resolutions, and analytical assumptions. Overall, the FW-H code has been shown to be able to predict these analytical sound sources with a sufficient accuracy.

The monopole validation have shown the numerical FW-H code's range of capability in predicting the thickness sources, using cases such as co-stationary, co-translation, and a translating source (Figures 2.7-2.11). The code was able to predict phenomena such as phase lagging, convective amplification, and Doppler shifting. The validation varied the discretisation resolutions between 275, 100, and 35 grid points per wavelength and 30, 15, and 5 time steps per period; with which the numerical FW-H achieved a decreasing errors between $\sim 0.5\%$ and $\sim 30\%$ using the finest and coarsest resolutions, respectively. This shows that the finest resolution is sufficient to properly predict these simple cases.

Alongside the monopole validation, the Retarded Time and Advanced Time algorithms have been successfully validated. Given a sufficient spatial and temporal discretisation resolutions, both algorithms achieved good agreements with the analytical solutions across all test cases. While the Retarded Time is more straightforward to implement and validate, the Advanced Time algorithm is more advantageous when applied to real CFD cases as it allows the use of the source's non-uniform time-stepping while being compatible to parallel computing [10, 32].

Moving to the dipole validation, while the direct analytical dipole derivations were limited, the numerical FW-H predicted the loading sources in good agreement with the respective assumption's analytical solutions, provided the test cases follow the assumption. The stationary assumption fails when the source is in motion, which is due to the assumption made in its derivation where $\partial r/\partial \tau = 0$, which causes the numerical FW-H to introduce a phase lag in its prediction (Figure 2.12). As shown in Equation 2.19, there is no Doppler factor $1 - M_r$ in any of the terms, which causes any motion to be invalid. On the other hand, the plane wave assumption is invalid when the permeable surface is placed in the near-field, which causes the numerical solution to over-predict the amplitudes while introducing a phase lag (Figure 2.13). This is because the velocity field is assumed to be in-phase with the pressure field, which is not the case in the acoustic near-field.

In contrast to the analytical derivations, the dual-monopole method is significantly more robust in all cases. By superimposing two equal monopoles of opposite signs separated at a distance of $k\delta \ll 1$ (k is the wave number), the code instead uses the analytical solution of the monopole, thereby omitting the assumptions made in the dipole velocity derivations. Even so, although the analytical dipole solutions were limited, they provided a good baseline against which the dual monopole approach was used to validate the loading noise. Additionally, the dual monopole approach allows the visualisation of the sound directivity, which is highly relevant in the following chapters to show how the sound waves are distributed in space.

To conclude, the FW-H method has been rigorously validated across different cases while simultaneously testing the discretisation resolution and time integration method. Validation strategies presented in this chapter are the first steps in building confidence in the FW-H code's noise prediction capability. The next chapter verifies the FW-H code's accuracy when used to mimic sound signatures coming from a known source as well as its sensitivity to varying Mach numbers, which is to understand how discretisation parameters affect its prediction when used in more complex cases.

Chapter 3

Sensitivity Analysis of the FW-H Method

3.1 Sensitivity Analysis Background and Methodology

With the FW-H code validated, the next step before implementing it into CFD data is establishing its sensitivity to spatial and temporal discretisation. While the moonopole validation cases touched on the effects of discretisation, this section defines the discretisation parameters that serve as the baseline for the subsequent sensitivity analysis.

3.1.1 Definition of Non-Dimensional Discretisation Parameters

XXXX MORE LITERATURE FOR THE DISCRETISATION RESOLUTIONS
In existing literatures [38, 5], discretisation is typically quantified relative to the wave properties rather than absolute physical dimensions. Spatial resolution is defined by the number of grid points per wavelength, and temporal resolution is defined by the number of time steps per wave period. Using the fundamental wave speed relationship,

$$c = \lambda f = \frac{\lambda}{T} \tag{3.1}$$

the spatial resolution parameter X is defined as the wavelength divided by the characteristic grid spacing. Similarly, the temporal resolution parameter P is defined as the wave period divided by the time-step size,

$$X = \frac{\lambda}{d} \quad \text{and} \quad P = \frac{T}{\Delta t} \tag{3.2}$$

where,

$$d = \frac{2\pi R}{N_{grid}} \quad \text{and} \quad \Delta t = \frac{T}{N_{time}}$$

It should be noted that these definitions are based on the freestream value of the wave parameters, i.e., not accounting for any Doppler shifting effects. As the wave compresses at high speeds, the required discretisation parameters need to be adjusted, and therefore, the values are higher than for stationary, freestream cases. The scaling factor to account for this is simply the Doppler factor, $1 - M$, which scales the resolution by the Mach number at which the source is moving. Therefore, when accounting for this effect, the grid and time-step sizing are redefined as,

$$N_{grid} = \frac{2\pi RX}{\lambda_s(1-M)} \quad \text{and} \quad N_{time} = \frac{T_s(1-M)}{P} \tag{3.3}$$

It is also important to note that these parameters are imperfect as they provide a nominal baseline rather than an exact local measurement. For example, when using an analytical sound source, the code uses a spherical domain constructed with a latitude-longitude grid, meaning the grid spacing are non-uniform across the longitude. Furthermore, when applying the Advanced Time formulation in CFD cases, the varying emission times project onto a non-uniform observer time.

In quantifying the code's sensitivity to the discretisation parameters, the same error quantification is used as the one used in Chapter 2, i.e., using the L2-norm to compare the instantaneous values of the analytical solution and the numerical integration results. After errors at a given P and X combinations are calculated, a 2D matrix is presented to provide the range of error at the given parameter ranges. This approach is selected as it gives a unifying approach to the error quantification as the validation methodology, as well as to provide an easily understood quantification of the code's sensitivity.

3.1.2 Background of Test Cases

The objective of this sensitivity analysis is to understand the FW-H code's behaviour across varying discretisation parameters before applying it to a real CFD case. Rotor noise is defined by two key characteristics. The first is periodicity, which is dominated by harmonics called at the Blade Passing Frequency (BPF) and its integer multiples. The second is the subsonic operating speed of the blades, which causes a Doppler shift that amplified the observed frequencies and pressure amplitudes while compressing the acoustic wavelength. The two test cases were developed to ensure the FW-H code correctly resolves these physical phenomena.

3.2 Multipole Ring Analysis

3.2.1 Background of Multipoles Analysis

To address the multi-harmonic characteristic of the propeller noise, a stationary ring of monopoles, referred to as the multipole ring case, is used. This setup replicates the thickness (monopole) noise data first established by Farassat [23] for a three bladed propeller, which remains a benchmark for research in aeroacoustics [24, 39]. Instead of modelling the blades, discrete point sources form a ring, with a radius equal to the reference propeller. This approach represents the "blade passing points," where the discrete sources represent thickness of the rotor blade tips passing through the ring's azimuthal positions.

3.2.2 Mimicking the Noise Signature of a Propeller using Analytical Multipoles

3.2.3 Sensitivity of the FW-H Method in Multipole Ring Prediction

3.3 Sensitivity Analysis of Rotating Monopoles

3.3.1 Background of Rotating Monopole Analysis

The second test approaches the second characteristic of the rotor noise by modelling a sound source in subsonic motion. Following the work by Ffowcs Williams [40], a single monopole point rotates at specific Mach numbers while emitting pressure waves. The radiation frequency (f_{source}) is defined as an integer multiple (n) of the rotational frequency (f_{rot}), such that $f_{source} = nf_{rot}$.

3.4 Concluding Summary

Chapter 4

Aerodynamic and Aeroacoustic Prediction of an Open Rotor using a Hybrid URANS-FW-H Method

Chapter 5

Conclusions

A Ffowcs Williams-Hawkings (FW-H) acoustic code has been analytically validated. The numerical integration is accurate, matching analytical derivations for stationary and moving monopole sources with less than 1% error under sufficient grid and temporal resolution.

For dipole sources, the explicit analytical derivations were shown to break down when subjected to non-zero relative motion or near-field control surfaces. To bypass these limitations, a dual-monopole approach was implemented, proving to be a good test case to demonstrate dipoles. While the numerical integration is mathematically sound, further refinement of the dipole analytical validation method is required. Once complete, the code will be integrated with URANS CFD for open rotor noise prediction.

Chapter 6

Future Work

With the FW-H solver validated with some limitations, the project moves to the hybrid analytical-computational framework. The next part of the work include:

- Validation refinements for dipoles (e.g., using non-stationary monopole as the source term, dynamic vector calculations) to reduce errors.
- Updating the numerical FW-H integration code to use double-precision floating-point arithmetic.
- Development and validation of a baseline URANS-FW-H hybrid model.
- Using the validated URANS-FW-H approach to predict open rotor noise.

Review of Generative AI Use

Generative AI (Google Gemini 3 Pro) has been used in augmenting the research and writing of this report. The usage of which are in helping with LaTeX formatting, formalising some English language uses, mathematical derivations, and checking for programming errors or bugs. The following are prompt examples used on the above main usages:

- ”Help me format this LaTeX document using the structure abstract, table of contents, nomenclature, intro, lit review, derivations, validations, discussion, conclusions, future work, appendices. The word count should only include the main content (from intro to future work).
- ”Help me rephrase these two sentences and bridge between them”
- ”In deriving the velocity equation for the dipole, as the explicit integral of the convolution between the right hand side term and Green’s function is unviable, what would be the implications of assuming that r is a constant value, such that some integrations of the F/(1-M_r) term can be done? If this is solved in the retarded time, the denominator is cancelled, so I can integrate F directly, correct?
- ”As the dual monopole method requires introducing an additional point source, which the code had not considered initially, help me to put an additional point monopole in the sample_properties subroutine in Fortran. As per the derivations, I need to separate the two by a distance delta, which I think can just be calculated as 0.01/k so it dynamically changes if any parameters are changed in the future.

The result of the first two examples being in the formatting of this report and some of the sentences within. None of the result sentences are directly copied from the tool, and instead are rephrased in the author’s own words to reflect the author’s understanding of the topic. The third prompt helped the author in deriving the mathematical parts presented in this report; of which the results are checked and verified via textbooks and during meetings with supervisors. The fourth prompt example returned Fortran codes that are used for the analytical solutions. These solutions tend to come with bugs which the author fixed independently.

In conclusion, all text-based outputs given by the Generative AI were checked, reworded, and edited to be based on the author’s understanding of the topic and to avoid plagiarism. Outputs that are technical are verified by textbooks and discussions with supervisors to ensure the accuracy of the results and limitations of the assumptions.

Bibliography

[1] N. Cumpsty and A. Heyes, Jet Propulsion. Cambridge University Press, 2015.

[2] S. Farokhi, Future propulsion systems and energy sources in sustainable aviation. John Wiley & Sons, Inc., 2020.

[3] D. B. Hanson, "Noise of counter-rotation propellers," in *AIAA*, vol. 22, pp. 609–617, 1985.

[4] J. M. Tyler and T. G. Sofrin, "Axial flow compressor noise studies," in *Pre-1964 SAE Technical Papers*, SAE International, Jan. 1962.

[5] M. H. Dunn, A. F. Tinetti, and D. M. Nark, "Open rotor noise prediction using the time domain formulations of farassat," tech. rep., 2014.

[6] E. Envia, "Contra-rotating open rotor tone noise prediction," tech. rep., 6 2014.

[7] *Prediction of the Aero-Acoustic Performance of Open Rotors*, vol. Volume 2A: Turbomachinery of *Turbo Expo*, 06 2014.

[8] A. P. Dowling and J. E. F. Williams, Sound and sources of sound. E. Horwood; Halsted Press, 1983.

[9] J. E. F. Williams and D. L. Hawkings, "Sound generation by turbulence and surfaces in arbitrary motion [321] sound generation by turbulence and surfaces in arbitfrary motion," *Source: Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences*, vol. 264, pp. 321–342, 1969.

[10] K. S. Brentner, "Numerical algorithms for acoustic integrals with examples for rotor noise prediction," *AIAA Journal*, vol. 35, pp. 625–630, 1997.

[11] A. Najafi-Yazdi, G. A. Brés, and L. Mongeau, "An acoustic analogy formulation for moving sources in uniformly moving media," *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 467, pp. 144–165, 1 2011.

[12] A. S. Morgans, *Transonic Helicopter Noise*. PhD thesis, University of Cambridge, 2004.

[13] S. K. Lele and J. W. Nichols, "A second golden age of aeroacoustics?," *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 372, 8 2014.

[14] G. Huang, A. Sharma, X. Chen, S. Jimeno, and A. Riaz, "Framework for multi-fidelity simulations of flow interaction and noise of an open rotor," in *AIAA Science and Technology Forum and Exposition, AIAA SciTech Forum 2025*, American Institute of Aeronautics and Astronautics Inc, AIAA, 2025.

[15] F. Farassat, M. H. Dunn, A. F. Tinetti, and D. M. Nark, "Open rotor noise prediction methods at nasa langley- a technology review," in *15th AIAA/CEAS Aeroacoustics Conference (30th AIAA Aeroacoustics Conference)*, 2009.

[16] W. Fan, J. Zhang, J. Shen, X. Sun, and L. Feng, "Verification of hybrid des-fw-h method for estimation of discharge noise induced by turbine cooler," in *Journal of Physics: Conference Series*, vol. 3004, Institute of Physics, 2025.

[17] *Prediction of Certification Noise Levels Generated by Contra-Rotating Open Rotor Engines*, 2012.

[18] P. H. Thomas, C. Hall, and A. S. Morgans, "Analytical solutions for acoustic integral solvers," in *18th AIAA/CEAS Aeroacoustics Conference (33rd AIAA Aeroacoustics Conference)*, American Institute of Aeronautics and Astronautics Inc., 2012.

[19] A. Sharma and H. N. Chen, "Prediction of aerodynamic tonal noise from open rotors," *Journal of Sound and Vibration*, vol. 332, pp. 3832–3845, 8 2013.

[20] *A Comparative Study of Computational Aeroacoustics Modelling of an Axial Fan Using CAA and FW-H Models in the URANS and DES Framework*, 2025.

[21] P. R. Spalart, "Detached-eddy simulation," *Annual Review of Fluid Mechanics*, vol. 41, pp. 181–202, 1 2009.

[22] J. E. Ffowcs, Williams, , and D. L. Hawkings, "Theory relating to the noise of rotating machinery," *J. Sound Vib*, p. 10, 1969.

[23] F. Farassat and T. Brown, "A new capability for predicting helicopter rotor and propeller noise including the effect of forward motion," tech. rep., National Aeronautics and Space Administration, 1977.

[24] C. Woan and G. Gregorek, *The exact numerical calculation of propeller noise*.

[25] A. B. Parry, *Theoretical Prediction of Counter-Rotating Propeller Noise*. PhD thesis, University of Leeds, 1988.

[26] T. Colonius and S. K. Lele, "Computational aeroacoustics: Progress on nonlinear problems of sound generation," 8 2004.

[27] J. J. P. M. C. D. B. Assistn and tt Professor, "Aerodynamic and acoustic calculations of transonic nonlifting hovering rotors," *Journal of the American Helicopter Society*, 1996.

[28] K. S. Brentner, A. S. Lyrintzis, and E. K. Koutsavdis, "A comparison of computational aeroacoustic prediction methods for transonic rotor noise," *Journal of Aircraft*, vol. 33, no. 4, pp. 771–779, 1996.

[29] K. S. Brentner and F. Farassat, "Modeling aerodynamically generated sound of helicopter rotors," *Progress in Aerospace Sciences*, vol. 39, pp. 83–120, 2003.

[30] A. Maaloum, S. Kouidri, and R. Rey, "Aeroacoustic performance evaluation of axial flow fans based on the unsteady pressure field on the blade surface," *Applied Acoustics*, vol. 65, pp. 367–384, 4 2004.

[31] P. D. Francescantonio, "A new boundary integral formulation for the prediction of sound radiation," *Journal of Sound and Vibration*, vol. 202, pp. 491–509, 7 1997.

[32] D. Casalino, "An advanced time approach for acoustic analogy predictions," *Journal of Sound and Vibration*, vol. 261, no. 4, pp. 583–612, 2003.

[33] A. Stuermer, J. Yin, and R. Akkermans, "Progress in aerodynamic and aeroacoustic integration of cror propulsion systems," *The Aeronautical Journal*, vol. 118, no. 1207, pp. 1137–1158, 2016.

[34] I. Lepot, M. Leborgne, R. Schnell, J. Yin, G. Delattre, F. Falissard, and J. Talbotec, "Aero-mechanical optimization of a contra-rotating open rotor and assessment of its aerodynamic and acoustic characteristics," *Proceedings of the Institution of Mechanical Engineers, Part A: Journal of Power and Energy*, vol. 225, no. 7, pp. 850–863, 2011.

[35] D. Lindblad, G. M. Villar, N. Andersson, A. C. Patrao, S.-k. Courty-Audren, and G. Napias, "Aeroacoustic analysis of a counter rotating open rotor based on the harmonic balance method," in *2018 AIAA Aerospace Sciences Meeting*, 2018.

[36] K. S. Brentner and F. Farassat, "Analytical comparison of the acoustic analogy and kirchhoff formulation for moving surfaces," *AIAA Journal*, vol. 36, pp. 1379–1386, 1998.

[37] Y. T. Char, F. Zhao, and A. Morgans, "Formulations 1a and 1c of the fw-h equation: Monopole test cases," tech. rep., Imperial College London, 2024.

[38] D. Lewis, C. Perrin, and F. Falissard, "Tonal noise predictions of a generic open-fan based on a hybrid urans approach," 05 2024.

[39] H. Wang, P. Shan, and Y. Zhou, "Development and application of open rotor discrete noise prediction program using time-domain methods," *Applied Sciences*, vol. 14, no. 3, 2024.

[40] D. G. Crighton, A. P. Dowling, J. E. Ffowcs Williams, M. Heckl, and F. G. Leppington, Modern Methods in Analytical Acoustics. Springer London, 1992.

Appendix A

Research Project Gantt Chart

The following is the Gantt Chart of the current project up to its completion. Some adjustments had to be made along the duration of the project's execution such as CFD-related parts, however, the overall project objectives and milestones remain the same.

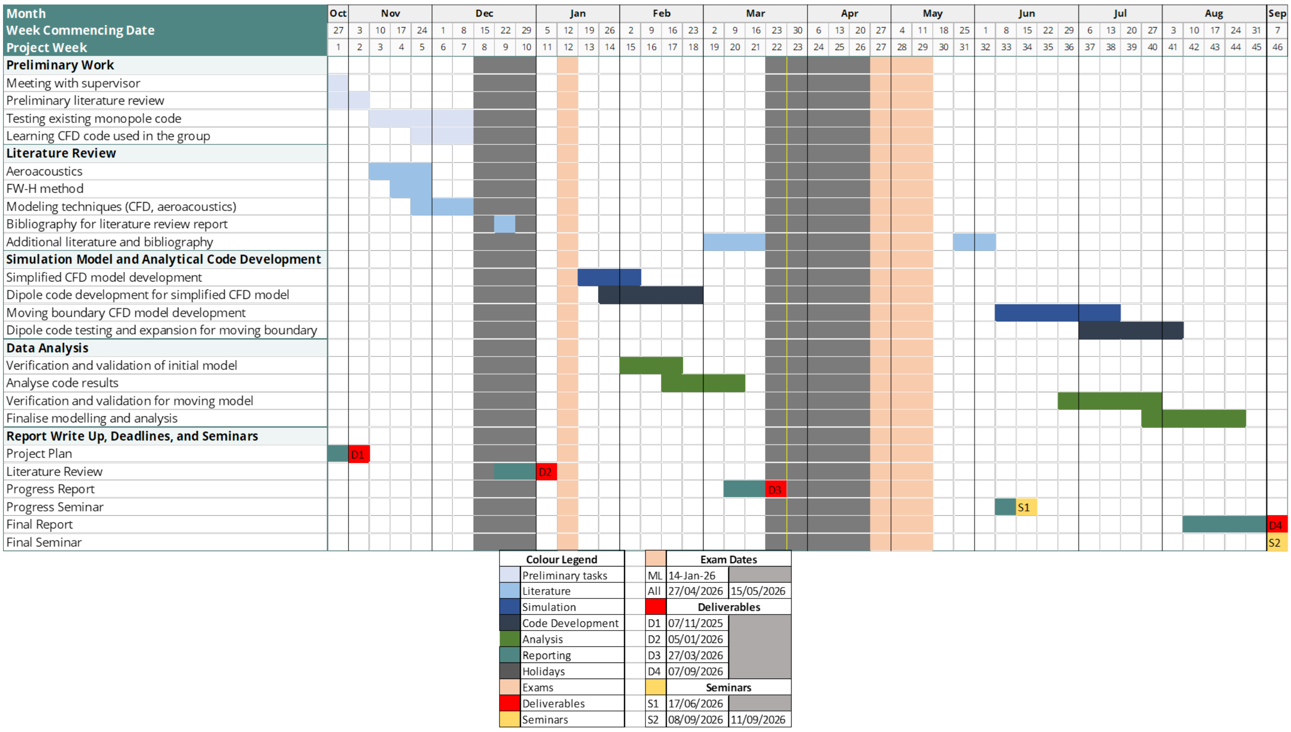


Figure A.1: Gantt Chart of the research project

Appendix B

Variable Derivatives

To derive the analytical point sources, a few definition for the derivatives are established.

$$\left.\frac{\partial r}{\partial \tau}\right|_x = -v_r \tag{B.1}$$

$$\left.\frac{\partial M_r}{\partial \tau}\right|_x = \left(\frac{\partial M}{\partial \tau}\right)_r + \frac{c(M_r^2 - |M|^2)}{r} \tag{B.2}$$

$$\left.\frac{\partial |1-M_r|}{\partial \tau}\right|_x = \frac{(1-M_r)}{|1-M_r|}\left(-\left(\frac{\partial M}{\partial \tau}\right)_r + \frac{c(|M|^2 - M_r^2)}{r}\right) \tag{B.3}$$

$$\left.\frac{\partial [Q(\tau)]_{\tau *}}{\partial \tau}\right|_x = \left.\left[\frac{1}{1-M_r}\frac{\partial Q(\tau)}{\partial \tau}\right]_x\right|_{\tau *} \tag{B.4}$$

$$-\left.\frac{\partial}{\partial x_i}\right|_t \int_S [Q_i]_{\tau *} dS = \left.\frac{\partial}{\partial t}\right|_x \int_S \left[\frac{Q_i r_i}{cr}\right]_{\tau *} dS + \int_S \left[\frac{Q_i r_i}{r^2}\right]_{\tau *} dS \tag{B.5}$$

Appendix C

Green's Function for Wave Equations

Given a wave equation,

$$\left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\psi(\hat{r},t) = -f(\hat{r},t)$$

Green's function of the above equation,

$$\left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)G(\hat{r},t|\hat{r'},t') = -\delta(\hat{r'}-\hat{r})\delta(t-t')$$

Let $\hat{r'} = 0$ and $t' = 0$. Solving for $\nabla^2 G$

$$\begin{aligned}
\nabla^2 G(\hat{r},t) &= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)G \\
&= \frac{1}{r}\frac{\partial G}{\partial r} + \frac{1}{r}\left(\frac{\partial G}{\partial r} + r\frac{\partial^2 G}{\partial r^2}\right) \\
&= \frac{1}{r}\frac{\partial^2}{\partial r^2}(rG).
\end{aligned}$$

Substituting back to the wave form,

$$\nabla^2 G - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}G = -\delta(r)\delta(t)$$

$$\frac{1}{r}\frac{\partial^2}{\partial r^2}(rG) - \frac{1}{c^2}\frac{1}{r}\frac{\partial^2}{\partial t^2}(rG) = -\delta(r)\delta(t)$$

Factoring the differential terms, it is found that if $r > 0$, then the left-hand side term is equal to zero,

$$\left[\left(\frac{1}{c}\frac{\partial}{\partial t} + \frac{\partial}{\partial r}\right)\left(\frac{1}{c}\frac{\partial}{\partial t} - \frac{\partial}{\partial r}\right)\right](rG) = 0 \tag{C.1}$$

Then, let $u = ct - r$ and $v = ct + r$, it will be found that,

$$\frac{\partial}{\partial u} = \frac{1}{2}\left(\frac{1}{c}\frac{\partial}{\partial t} - \frac{\partial}{\partial r}\right) \quad \frac{\partial}{\partial v} = \frac{1}{2}\left(\frac{1}{c}\frac{\partial}{\partial t} + \frac{\partial}{\partial r}\right)$$

Given that $\partial^2(rG)/(\partial u \partial v) = 0$, if $rG = g(v) = g_+(ct + r)$ and $rG = g(u) = g_-(ct - r)$, then,

$$G = \frac{1}{r} g_\pm(ct \pm r)$$
$$G(\hat{r}, t|\hat{r'}, t') = \frac{1}{\hat{r} - \hat{r'}} g_\pm(c(t - t') \pm |\hat{r} - \hat{r'}|)$$

Taking $\nabla^2 G$ using this definition,

$$\nabla^2 G = \nabla^2 \left(\frac{1}{r} g_\pm(ct \pm r)\right) = \nabla^2 \left(\frac{1}{r}\nabla g_\pm + g_\pm \nabla\left(\frac{1}{r}\right)\right)$$
$$= -\frac{2}{r^2}\frac{\partial g}{\partial r} + \frac{2}{r^2}\frac{\partial g}{\partial r} + \frac{1}{r}\frac{\partial^2 g}{\partial r^2} - 4\pi\delta(r)g$$

The first two terms of the right-hand side cancel, and moving back to the wave equation format,

$$\nabla^2 G - \frac{1}{c^2}\frac{\partial^2}{\partial t^2} G = -\frac{1}{r}\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial r^2}\right) g_\pm - 4\pi\delta(r) g_\pm \tag{C.2}$$

As described in equation C.1, the first term of the right-hand side cancels. Going back to the wave equation,

$$\Box^2 \psi(\hat{r}, t) = -\delta(\hat{r})\delta(t) = -4\pi\delta(\hat{r}) g_\pm$$

Therefore,

$$\therefore \quad G_\pm = \frac{\delta(ct \pm r)}{4\pi r} \tag{C.3}$$

Appendix D

Replacing $\partial/\partial x_i$ Terms

As the definition of dipoles are that of the spatial derivative of the forcing term convolved with the Green's function in 3D free-space, it is necessary to replace the spatial derivative terms to that of the temporal one to derive using the definitions described in Appendix B. As the goal is to transform $\mathbf{x} \rightarrow \mathbf{X}$ and $t \rightarrow \tau$, and the following are true: $\mathbf{x} = \mathbf{X}$ and $\tau = t - \frac{\mathbf{r}}{c}$, then

$$\left.\frac{\partial}{\partial x_i}\right|_t = \frac{\partial X_j}{\partial x_i}\frac{\partial}{\partial X_j} + \frac{\partial \tau}{\partial x_i}\frac{\partial}{\partial \tau} = \delta_{ij}\frac{\partial}{\partial X_j} - \frac{r_i}{cr(1-M_r)}\frac{\partial}{\partial \tau}$$

$$\frac{\partial}{\partial x_i} = \frac{\partial}{\partial x_i} - \frac{r_i}{cr(1-M_r)}\frac{\partial}{\partial \tau} \tag{D.1}$$

Similarly for the time derivative,

$$\left.\frac{\partial}{\partial t}\right|_{\mathbf{x}} = \frac{\partial X_j}{\partial t}\frac{\partial}{\partial X_j} + \frac{\partial \tau}{\partial t}\frac{\partial}{\partial \tau}$$

Assuming no mean flow, the first term of the right hand side, $\partial X_i/\partial t$ is equal to zero, leaving,

$$\left.\frac{\partial}{\partial t}\right|_{\mathbf{x}} = \frac{1}{1-M_r}\left.\frac{\partial}{\partial \tau}\right|_{\mathbf{x}} \tag{D.2}$$

Appendix E

Monopole Velocity Derivations

Using the momentum equation, assuming the flow has no mean flow and is inviscid, substituting the monopole definition into the pressure term, the second terms of both left- and right-hand are zero. Defining the pressure gradient by defining the pressure as monopole pressure definition,

$$-\frac{\partial p}{\partial x_i} = \frac{\partial}{\partial x_i}\frac{\partial}{\partial t}\left[\frac{Q(\tau)}{4\pi r(1-M_r)}\right]_{\tau*}$$

and cancelling the unsteady term from both sides gives the following for the momentum equation,

$$u_i = \frac{1}{\rho_0}\frac{\partial}{\partial x_i}\left[\frac{Q(\tau)}{4\pi r(1-M_r)}\right]_{\tau*} \tag{E.1}$$

Convolving with Green's function in free 3D-space is done to replace the spatial derivative into a temporal derivative,

$$p'(\boldsymbol{x},t) = \frac{\partial}{\partial x_i}(Q_i \star G) = -\frac{\partial}{\partial t}\left(Q_i \star \frac{x_i G}{c|\boldsymbol{x}|}\right) - Q_i \star \frac{x_i G}{|\boldsymbol{x}|^2}$$

which has the same form as the integral shown in Appendix B, Equation B.5. Substituting the monopole source terms into Q_i, and converting to retarded time using the equation found in Appendix D Equation D.2, gives,

$$\frac{\partial}{\partial x_i}\left[\frac{Q(\tau)}{4\pi r(1-M_r)}\right]_{\tau*} = \frac{1}{1-M_r}\frac{\partial}{\partial \tau}\left[\frac{Q(\tau)r_i}{4\pi c\boldsymbol{r}^2(1-M_r)^2}\right]_{\tau*} + \left[\frac{Q(\tau)r_i}{4\pi r^3(1-M_r)}\right]_{\tau*}$$

which can then be solved analytically to give the solution for velocity,

$$u_i = \frac{1}{4\pi\rho_0}\left[\frac{(\partial Q(\tau)/\partial\tau)r_i}{cr^2(1-M_r)^2} + \frac{Q(\tau)r_i}{r^3(1-M_r)^3} + \left(\frac{dM}{d\tau}\right)\frac{Q(\tau)r_i}{cr^2(1-M_r)^3} - \frac{Q(\tau)M^2 r_i}{r^3(1-M_r)^3} - \frac{Q(\tau)M_i}{r^2(1-M_r)^2}\right]_{\tau*} \tag{E.2}$$

Appendix F

Dipole Velocity: Stationary Assumption

Defining pressure gradient from the momentum equation and dipole definition,

$$-\frac{\partial p}{\partial x_i} = \frac{\partial^2}{\partial x_i \partial x_j}\left[\frac{f_i(\tau)}{4\pi r(1-M_r)}\right]_{\tau *}$$

Going back to momentum equation, the velocity is,

$$u_i = \frac{1}{4\pi\rho_0}\frac{\partial^2}{\partial x_i \partial x_j}\int_{-\infty}^{t}\frac{f_i(\tau)}{r(1-M_r)}dt$$

assuming $dr/dt = 0$ and $M_r = 0$, letting $F_i(\tau) = \int f_i(\tau)d\tau \implies f_i(\tau) = \dot{F}_i(\tau)$, and changing to integrating with respect to τ, this simplifies the integrand to,

$$u_i = \frac{1}{4\pi r\rho_0}\frac{\partial^2}{\partial x_i \partial x_j}\int_{-\infty}^{t} f_i(\tau)d\tau$$

Taking the integral,

$$u_i = \frac{1}{4\pi r\rho_0}\frac{\partial^2}{\partial x_i \partial x_j}\int_{-\infty}^{t} f_i(\tau)d\tau = \frac{1}{4\pi r\rho_0}\frac{\partial^2}{\partial x_i \partial x_j}\left[F_i(\tau - r/c)\right]$$

Hence, the analytical solution for velocity that is valid for cases of zero relative Mach number between the source and observer for near- and far-field,

$$u_i = \frac{1}{4\pi\rho_0}\left(\frac{F_j(3n_in_j - \delta_{ij})}{r^3} + \frac{f_j(3n_in_j - \delta_{ij})}{cr^2} + \frac{\dot{f}_j(n_in_j)}{c^2 r}\right) \tag{F.1}$$

Appendix G

Dipole Velocity: Plane Wave Assumption

Given the plane wave,

$$p'(\boldsymbol{x}, t) = \frac{g(t - r/c)}{r} \tag{G.1}$$

Taking the pressure gradient,

$$\frac{\partial p'}{\partial x_i} = \left[\frac{\partial}{\partial x_i}\frac{1}{r}g\right] + \frac{1}{r}\left[\frac{\partial}{\partial x_i}g\right]$$

where the first and second terms correspond to the near- and far-field, respectively. The plane wave is valid only in the far-field, and taking the time derivative yields,

$$\frac{\partial p'}{\partial t} = \left[\frac{\partial}{\partial t}\frac{g}{r}\right] = \frac{1}{r}\frac{\partial}{\partial t}g(\tau) - g(\tau)\frac{v_r}{r^2}$$

In the far-field, $\frac{1}{r}\frac{\partial}{\partial t}g(\tau) \gg g(\tau)\frac{v_r}{r^2}$, simplifying the expression to,

$$\frac{\partial p'}{\partial t} \approx \frac{1}{r}\frac{\partial}{\partial t}g(\tau)$$

which gives the following expression,

$$\frac{\partial p'}{\partial x_i} = -\frac{n_i}{c}\frac{\partial p'}{\partial t}$$

substituting into the momentum equation provides the velocity equation for the plane wave,

$$u_i = \frac{n_i}{\rho_0 c}p'(\boldsymbol{x}, t) \tag{G.2}$$